

Design and control of Dual Active Bridge

Descripción

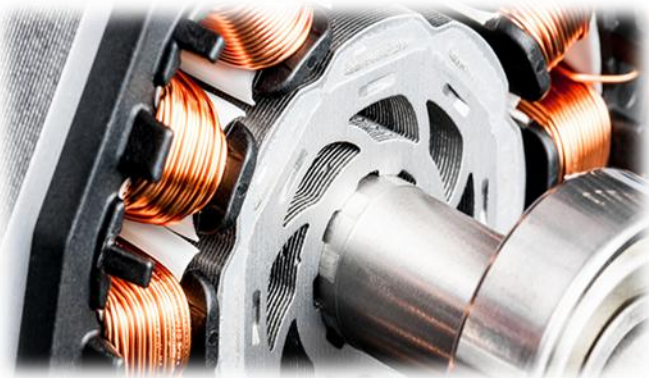


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Power and Energy Conversion Laboratory

Research in Power Electronics



Lecture Sections **Dual Active Bridge (DAB)**

1. Presentation of the dual active bridge (DAB) and soft switching
2. Modulations and nomenclature consensus
3. Single phase shift (SPS) analytical modeling
4. Conditions for Soft Switching with SPS
5. DAB control strategies with SPS
6. Analytical modeling with Triple Phase Shift (TPS)
7. Conditions for Soft Switching with TPS
8. GHM modeling with Triple Phase Shift
9. DAB control strategies with TPS
10. Optimal DAB design



Compilation of base papers

- Following a thorough search on the IEEE Xplore website, 23 prestigious papers studying DAB in its various fields have been shortlisted.
- Of these, 7 have been selected as a basis. They are used both to present explicit information and to extract concepts and generate our own developments.



- 1) 2014 Overview of Dual-Active-Bridge Isolated Bidirectional DC–DC Converter for High-Frequency-Link Power-Conversion System
- 2) 2014 Different Purpose Design Strategies and Techniques to Improve the Performance of a Dual Active Bridge With Phase-Shift Control
- 3) 2018c Analytical Small-Signal Transfer Functions for Phase Shift Modulated Dual Active Bridge Converters Using Phasor Transformation
- 4) 2017c Dual active bridge with triple phase shift by obtaining soft switching in all operating range
- 5) 2019 Overview and Comparison of Modulation and Control Strategies for a Nonresonant Single-Phase Dual-Active-Bridge DC–DC Converter
- 6) 2022 RMS Current Minimization in a SiC-Based Dual Active Bridge Converter Using Triple-Phase-Shift Modulation
- 7) 2020 Optimal Design of a Dual-Active-Bridge DC–DC Converter

Compilation of base papers



Content covered in the selected papers:

- [1] Overview of SPS, EPS, DPS, and TPS modulation types. Different topologies for the DAB AC link. Comments on ZVS. Application examples, construction features, and design recommendations.
- [2] Analytical DAB modeling with SPS. Small signal model for voltage control as a first-order plant. ZVS conditions and ZVS band in primary and secondary switches. Explanation of angular excursion reduction by decreasing equivalent inductance to improve linear power transfer behavior.
- [3] Analytical derivation of a small-signal model based on a phasor approximation of the DAB with SPS. A transfer function is obtained for current control with a source at the secondary port and a function for voltage control with a resistive load at the secondary port.
- [4] Analytical study of the DAB with TPS. Obtaining conditions for ZVS and ZCS in general for switching modes in the DAB operating range. A relationship is found that optimizes the soft switching range in the switching mode where it is possible for the DAB to have Soft Switching throughout its range.
- [5] Overview of DAB modulation and control. Generalization and improved phase shift modulation figures. Description of multiple operating modes with TPS. Operation with ZVS and ZCS. Control scheme with LUT based on optimization.
- [6] Obtaining operating values for the DAB with TPS through optimization using general harmonic approximation modeling. Clear approach to modeling and optimization to minimize rms squared currents in order to minimize conduction losses and using power constraints to establish energy flow. Soft switching is not taken into account for the analysis.
- [7] Presents an optimal design methodology for the construction and operation of a variable range DAB with TPS. Considers the optimization of rms current and soft switching conditions. Provides design criteria and an experimental example.

Section 1: Topology

Dual Active Bridge



DAB – Topology

The Dual Active Bridge (DAB) is an isolated, bidirectional DC–DC converter widely used in applications requiring bidirectional power transfer, high efficiency, and galvanic isolation [1, 5, 7]. Its basic structure consists of:

- Two complete bridges (H-bridge): one on the primary side and one on the secondary side.
- A high-frequency transformer provides isolation and voltage level matching.
- A series inductor: generally, the leakage inductance of the transformer plus an external inductance, which controls energy transfer.

The operating principle is based on applying square waveforms to both bridges and shifting them in time to control the transferred power. This shift generates a voltage difference in the inductor, which produces the energy flow between the two DC ports. In addition, it is possible to modify the duty cycle of the square waves of each AC port to achieve improvements in the converter's operating range.

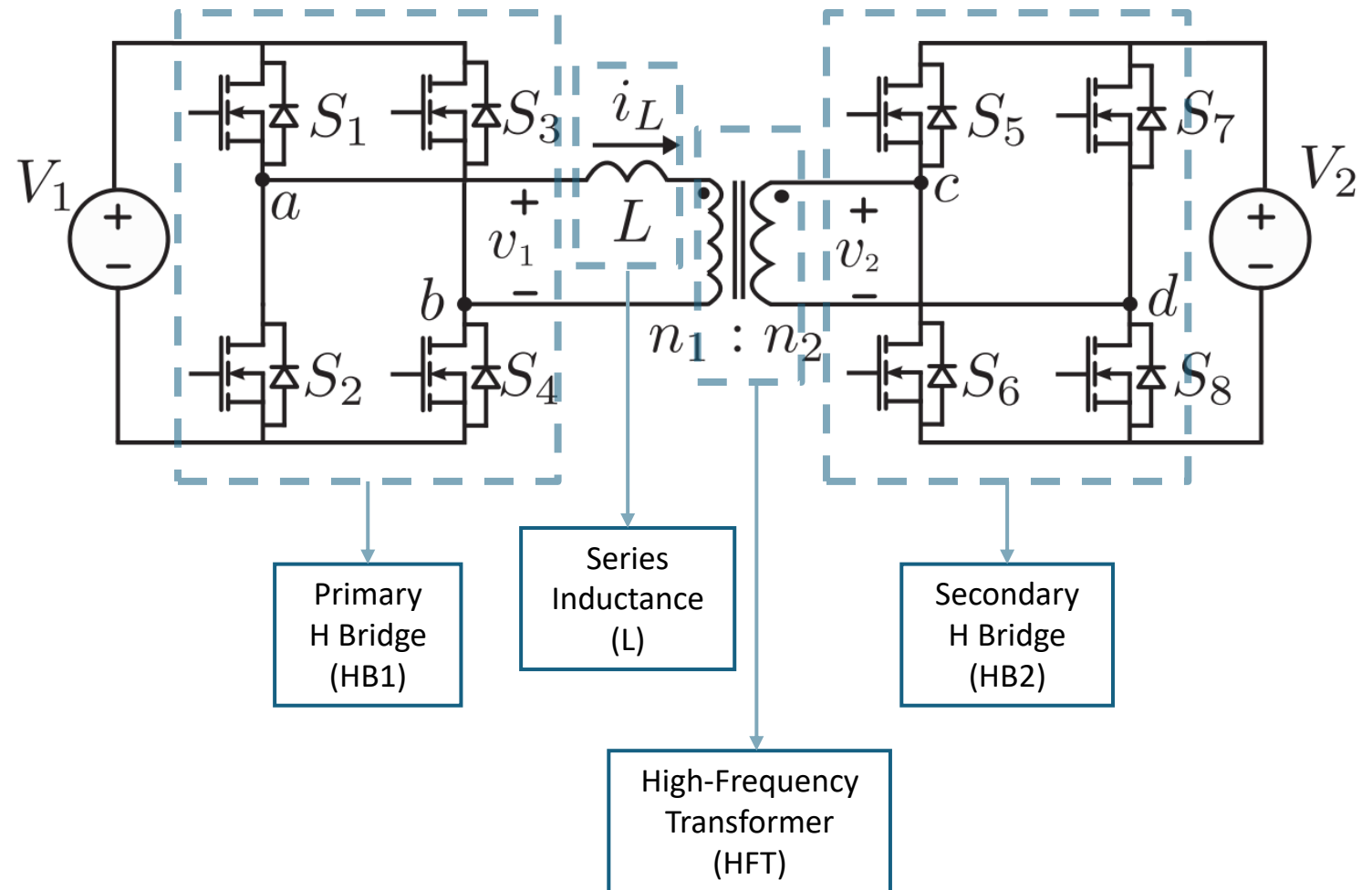


Fig. 1.1. Dual Active Bridge topology with marked parts. Based on [7]

DAB – Characteristics [1,5,7]

- Isolation: Galvanic, via transformer.
- Bidirectionality: Ability to transfer energy in both directions.
- Phase shift control: Adjustment of the angle between square waveforms.
- Control flexibility: Power flow can be controlled using low, medium, and high complexity strategies.
- Advanced modulation (TPS): Three degrees of freedom for optimization.
- Soft switching: ZVS and ZCS possible across the entire range with TPS.
- High efficiency: >95% in optimized designs.
- High power density: Operation at typical frequencies of 20–100 kHz.
- Applications: EV, BESS, SST, microgrids.
- Critical design: Optimal selection of transformation ratio and inductance to minimize RMS current and losses.
- Recommended components: SiC/GaN semiconductors for high frequency and low loss.

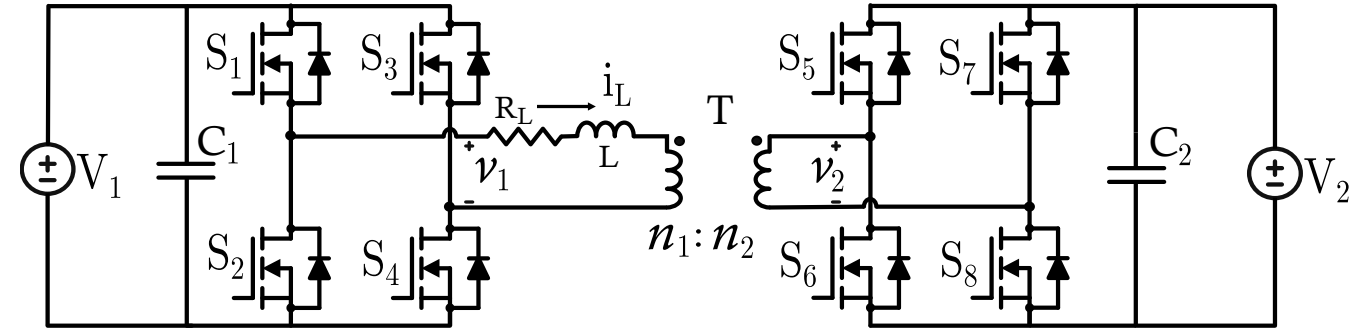


Fig. 1.2. Dual Active Non-Resonant Bridge Topology. Based on [1]

The principle of operation can be understood using the analogy of power transfer between two sinusoidal alternating networks, which depends on the relative phase shift φ between the voltage signals (1.1).

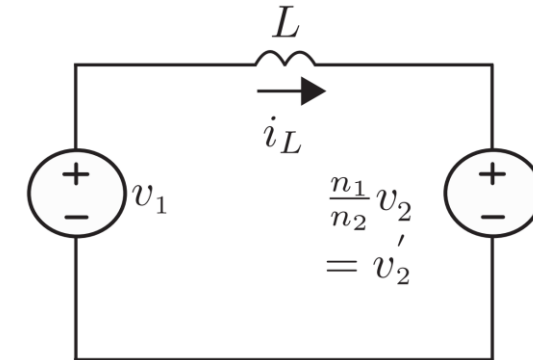


Fig. 1.3. Simple sinusoidal AC transfer circuit

$$P_{sine} = \frac{V_{1rms} V'_{2rms}}{2\pi f_s L} \sin(\varphi) \quad (1.1)$$

Soft Switching

Soft switching is a technique used in power electronics to improve efficiency and reduce stress on devices during switching [4-8]. The goal is to prevent switching from occurring with high voltage and high current simultaneously, giving rise to two types of soft switching:

1. ZVS (Zero Voltage Switching)

- It consists of activating the switching device to turn it on (Turn On) when there is zero or almost zero voltage at its terminals.
- Reduces EMI generation.

2. ZCS (Zero Current Switching)

- It consists of activating the switching device to deactivate its conduction (Turn Off) when the current it is conducting is zero or almost zero.
- Reduces thermal stress on devices.

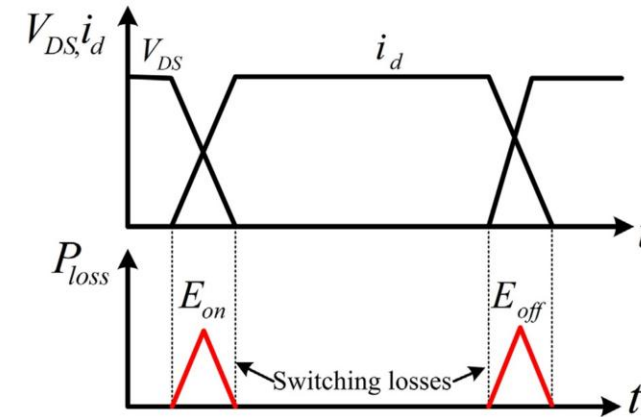


Fig. 1.4. Hard Switching [8]

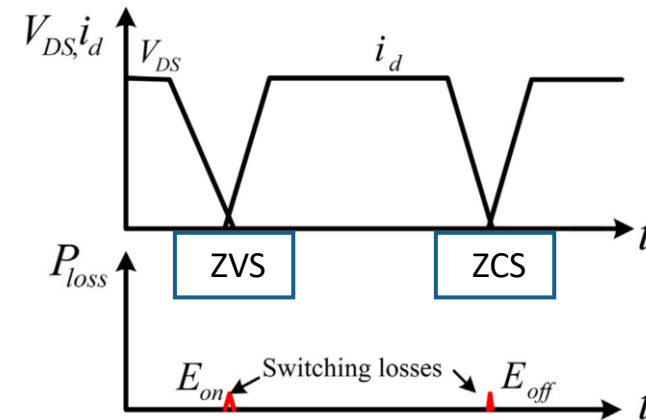


Fig. 1.5. Soft Switching [8]

Section 2: Modulations

Dual Active Bridge



DAB – Modulations

The operating principle is based on applying square wave signals to the two bridges and adjusting their timing to control the power transferred. There are several modulation strategies that vary in complexity and performance. [1, 5]

- **SPS (Single Phase Shift):** Single phase shift between bridges.
- **EPS (Extended Phase Shift):** Adds a second internal phase shift to one of the bridges, i.e., modifies one duty cycle.
- **DPS (Dual Phase Shift):** Introduces internal phase shifts to both bridges, i.e., modifies the duty cycles of both bridges equally.
- **TPS (Triple Phase Shift):** More advanced control with three degrees of freedom (phase shift between bridges and the duty cycle of each).

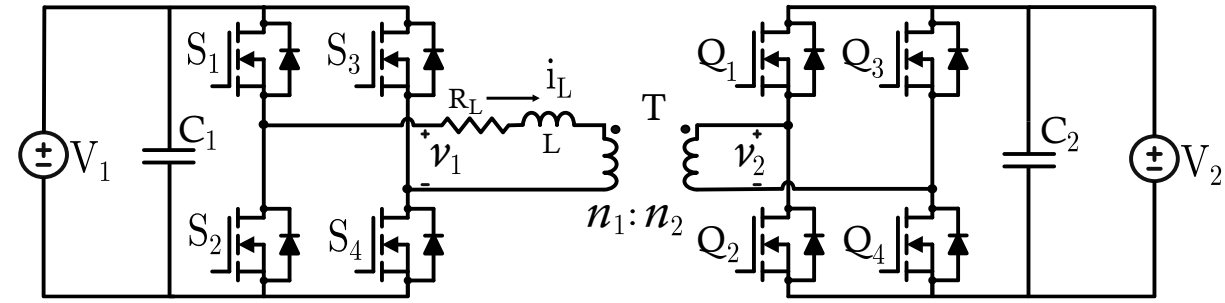


Fig. 2.1. Dual Active Non-Resonant Bridge Topology. Based on [1]

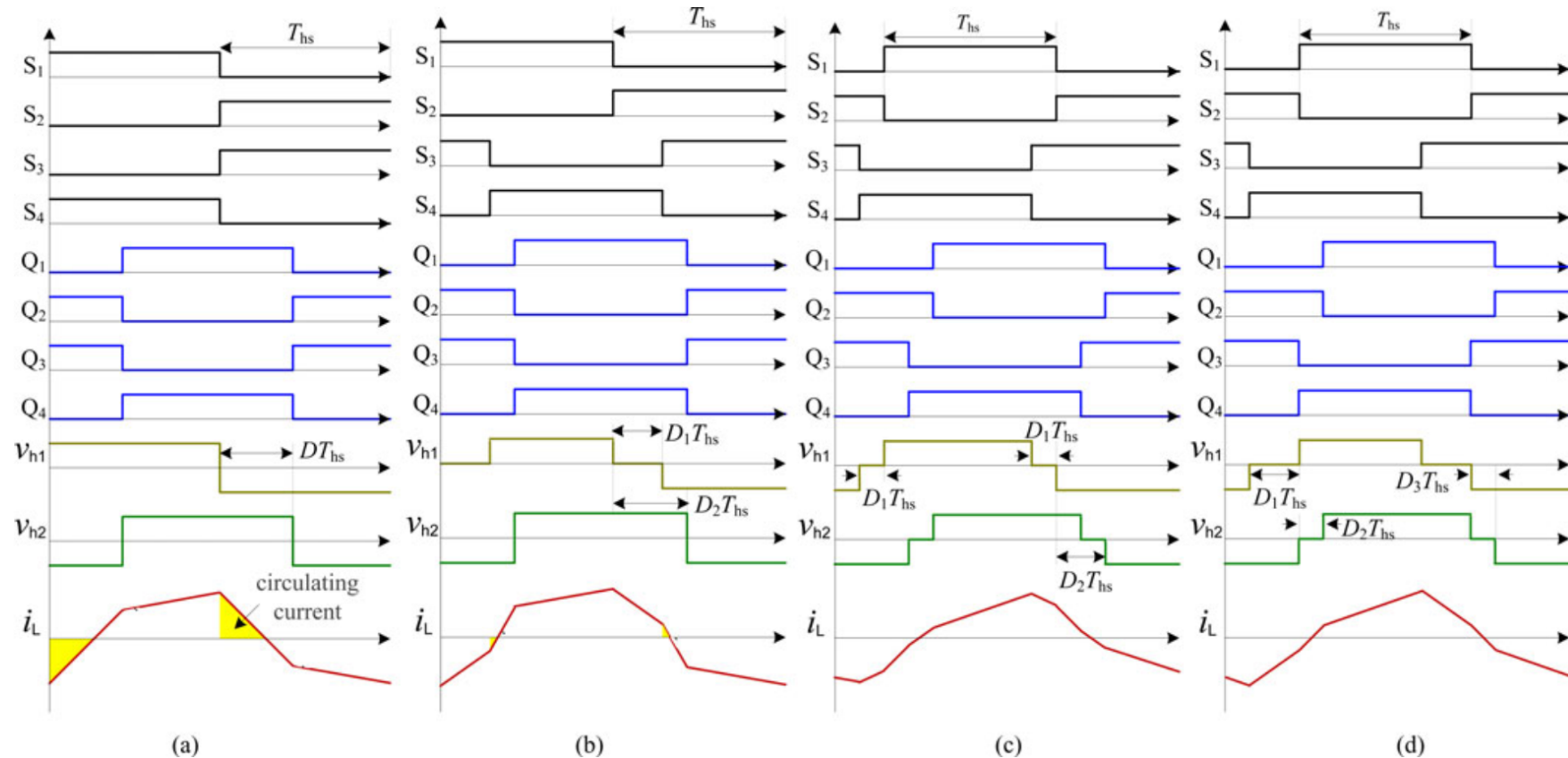


Fig. 2.2. Modulations based on phase shift in DAB.

(a) Single Phase Shift. (b) Extended Phase Shift (c) Dual Phase Shift. (c) Triple Phase Shift. [1].

DAB – Modulations

- TPS considers the independent modification of the duty cycle of the voltage-modulated square wave in bridge 1, the duty cycle in bridge 2, and the phase shift between both voltages. Therefore, SPS, EPS, and DPS can be seen as special cases of TPS [1, 5].

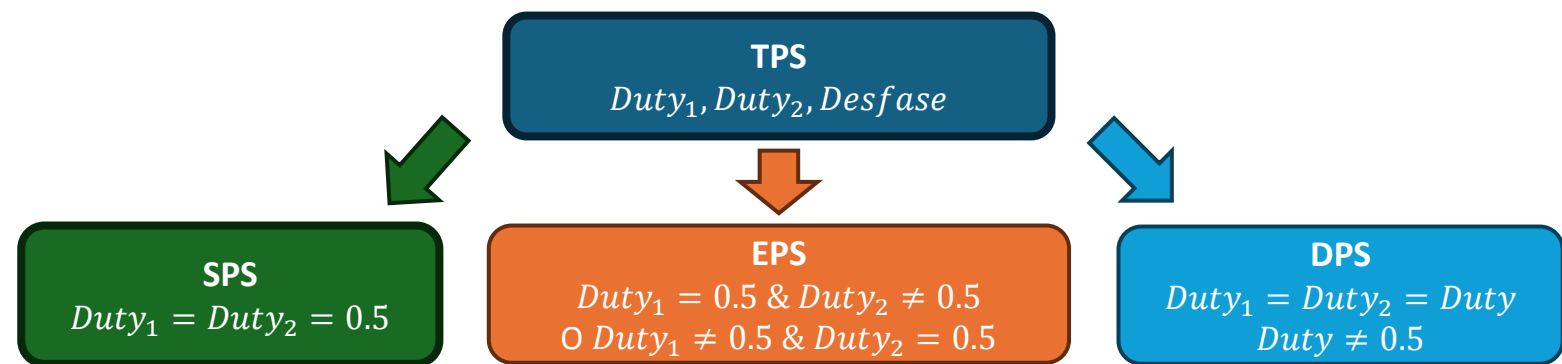


Fig. 2.3. Modulation similarity diagram

Table 2.1. Advantages and disadvantages of modulations

Modulation	Degrees of freedom	Advantages	Disadvantages
SPS (Single Phase Shift)	1 (phase shift between bridges)	- Simplicity. - Good overall dynamic response.	- High RMS current at low load. - Limited ZVS range. - Low efficiency outside nominal point.
EPS (Extended Phase Shift)	2 (external phase + internal phase shift in a bridge).	- Reduces circulating current. - Extends ZVS range. - Improves efficiency at partial load.	- Greater complexity than SPS.. - Requires status exchange between bridges according to power direction.
DPS (Dual Phase Shift)	2 (external phase shift + internal phase shift in both bridges).	- Reduces RMS current and stress on components. - Extends ZVS range more than EPS. - More uniform dynamics in power direction changes.	- Medium complexity. - Does not guarantee ZVS across the entire range.
TPS (Triple Phase Shift)	3 (two duty cycles + phase shift between bridges).	- Enables full optimization (minimum RMS current, reduced losses). - Guarantees ZVS and even ZCS across the entire operating range.. - Ideal for high-efficiency applications and wide voltage ranges.	- High control complexity. - Requires optimization algorithms and greater processing capacity.

Consensus on Nomenclatures

- Because in the vast documented research on DAB, multiple authors use different nomenclatures or ways to name control variables (duty cycles and bridge offset), it is necessary to choose one as a basis for the study.
- For all items used [1-7], the equivalence with the chosen base nomenclature will be detailed, which is the one used in [6], where:
 - $D_1 \in [0, 0.5]$: primary H-bridge duty cycle.
 - $D_2 \in [0, 0.5]$: secondary H-bridge duty cycle.
 - $\varphi \in [-2\pi, 2\pi]$: angular phase shift between bridges.
 - $d_{12} \text{ o } \Delta = \varphi/2\pi \in [-1, 1]$: temporal phase shift.
 - $N = a_{12} = n_1/n_2$: transformer turns ratio.
- ❖ $D_i = 0$ means that the i bridge synthesizes $v_i = 0$ [V]; $D_i = 0.5$ means that the bridge modulates a symmetrical 2-level square wave signal with 50% +Vdc and 50% -Vdc.
- ❖ $\varphi = 0 = 2\pi$ [rad] means that the voltage signals v_1 and v_2 are in phase when measured between the midpoints of their half-cycles, equivalent to measuring the phase difference between the peaks of two sinusoidal signals.
- ❖ Example: $\varphi = \pi/2$ [rad] equals to $d_{12} = \Delta = 0.25$, that is, 25% of T_s that bridge 1 leads bridge 2, with energy flowing from 1 to 2.

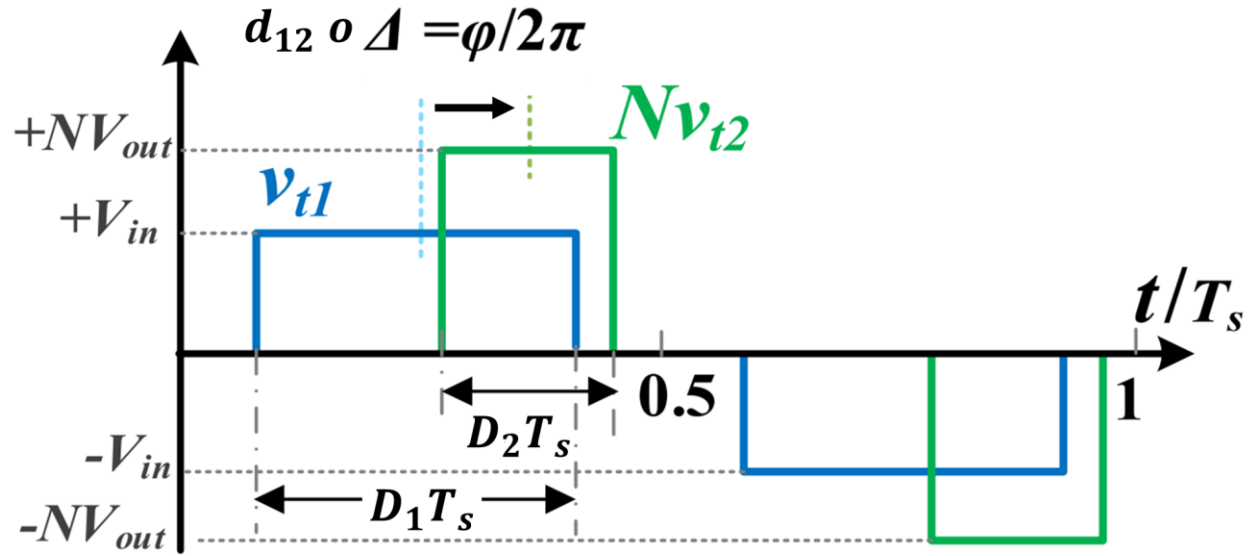


Fig. 2.4. Waveform of voltages at AC ports with TPS. Based on [6]

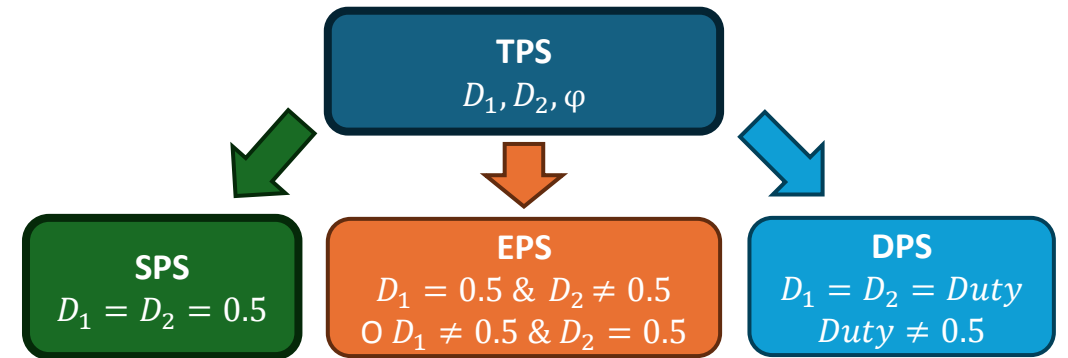


Fig. 2.5. Reduced modulation similarity diagram

DAB – Modulations to study

- We will delve deeper into DAB, operating with the simplest modulation based on the basic SPS operating principle and then with TPS.

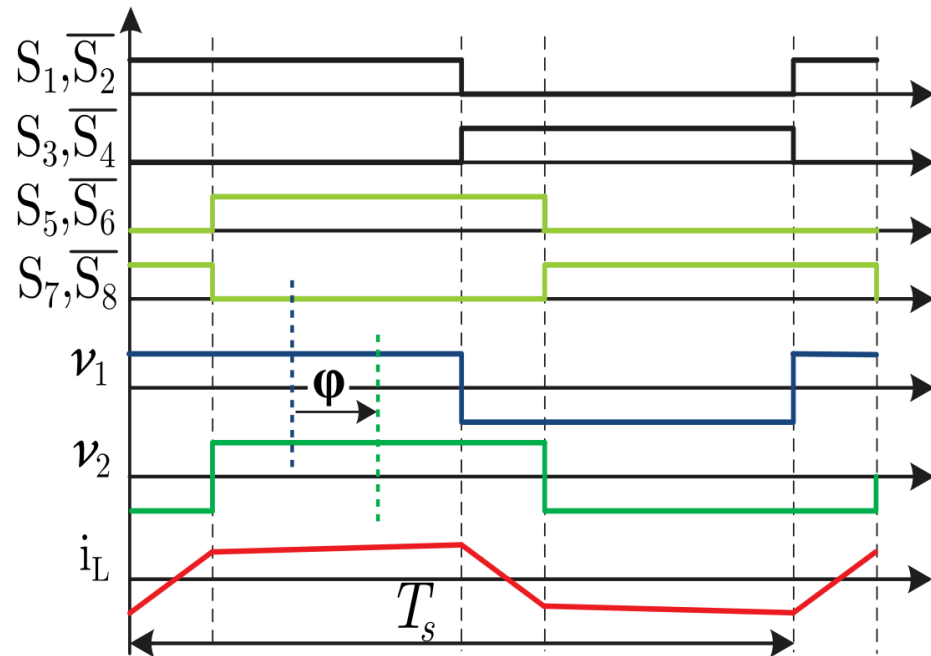


Fig. 2.7. Waveforms with SPS. Based on [5] and [6]

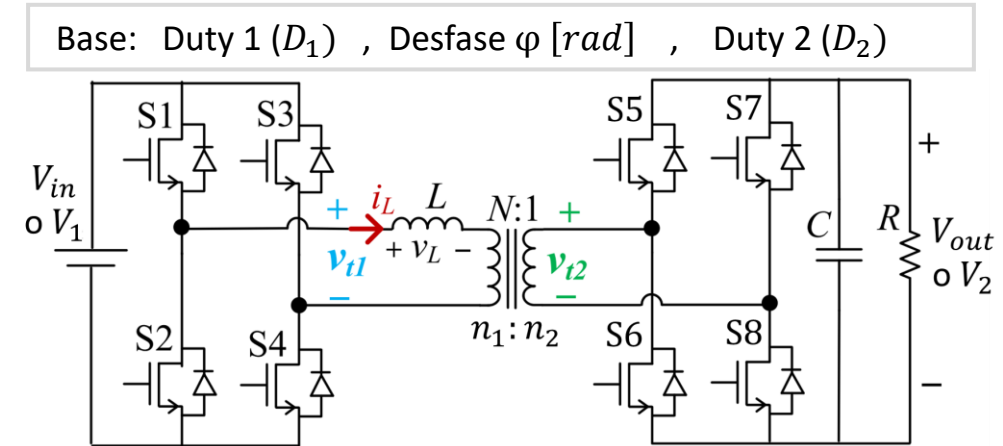


Fig. 2.6. Dual Active Non-Resonant Bridge Topology. Based on [6]

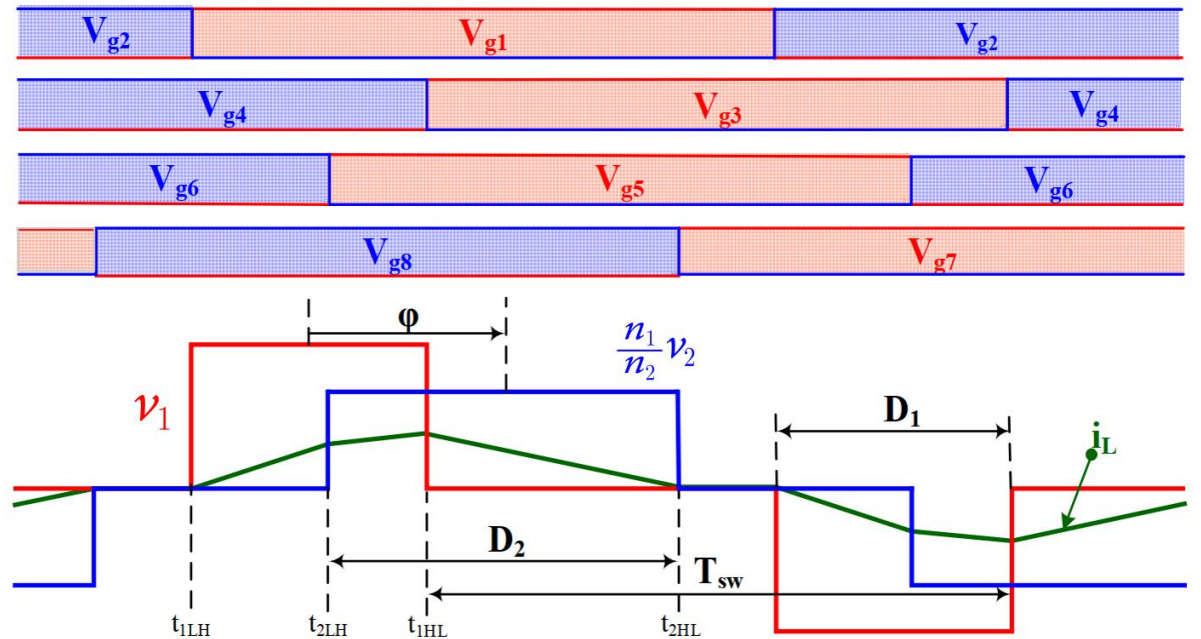


Fig. 2.8. Waveforms with TPS. Based on [4]

Section 3: Analytical modeling with SPS

Dual Active Bridge



DAB – SPS modeling

- A summary of the analytical modeling of DAB with SPS modulation can be found in [2], where DAB is presented in Fig. 3.2, and the waveforms in Fig. 3.4.
- The equivalence of the base nomenclature [b] with the variables of the article [2] is:
 - ❖ Turn ratio : $N = n_1/n_2 [b] = 1/n [2]$
 - ❖ Temporal phase shift : $d_{12} = \varphi/2\pi [b] = d/2 [2]$

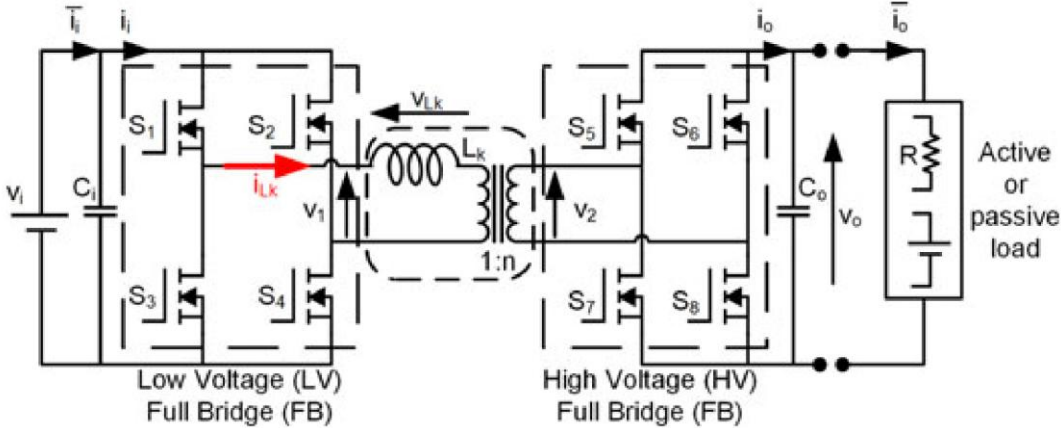


Fig. 3.2. Dual Active Bridge Topology in [2]

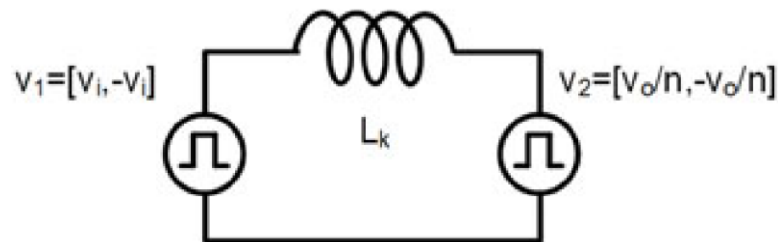


Fig. 3.3. Simplified Dual Active Bridge circuit in [2]

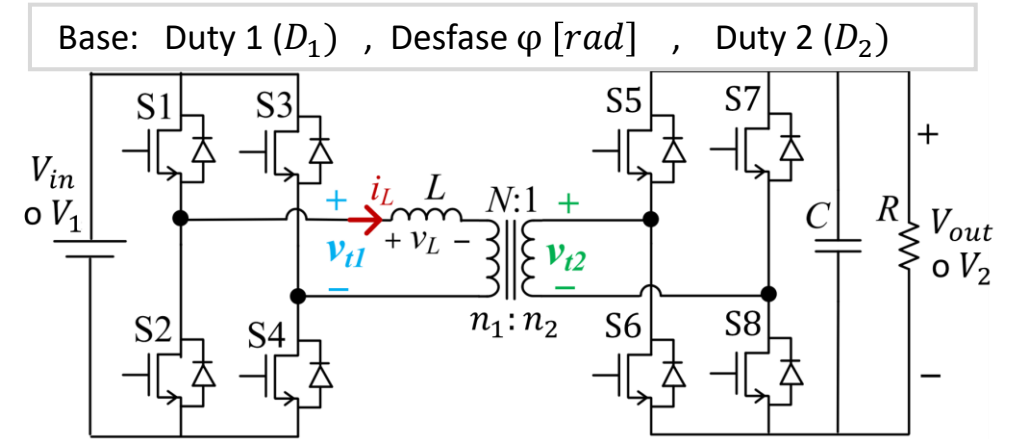


Fig. 3.1. Dual Active Non-Resonant Bridge Topology with base nomenclature [6]

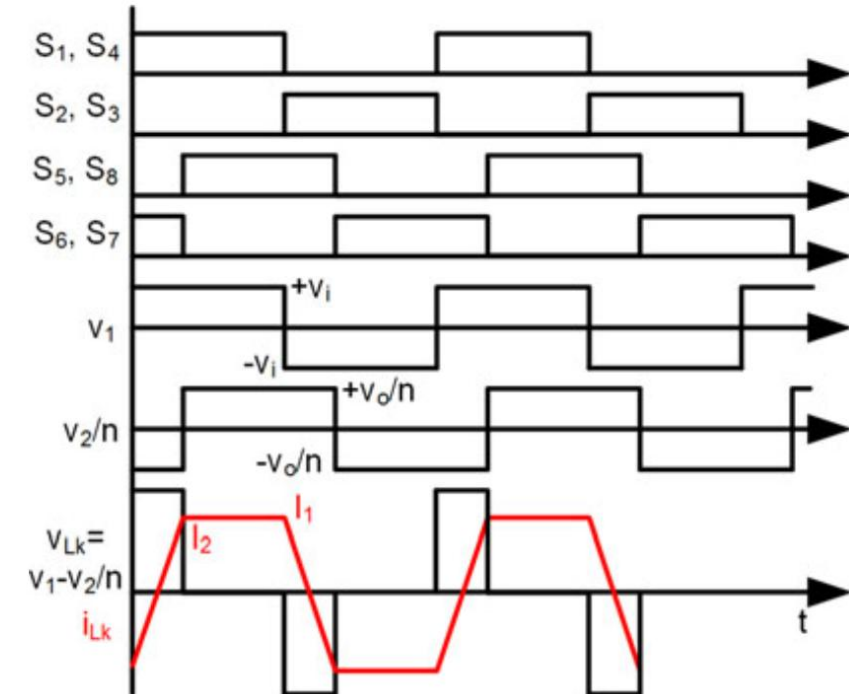


Fig. 3.4. DAB waveforms with SPS in [2]

DAB – SPS modeling [2]

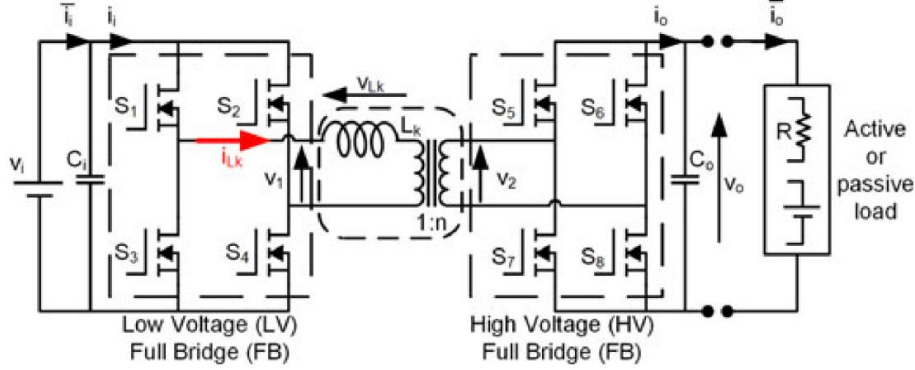


Fig. 3.2. Dual Active Bridge Topology in [2]

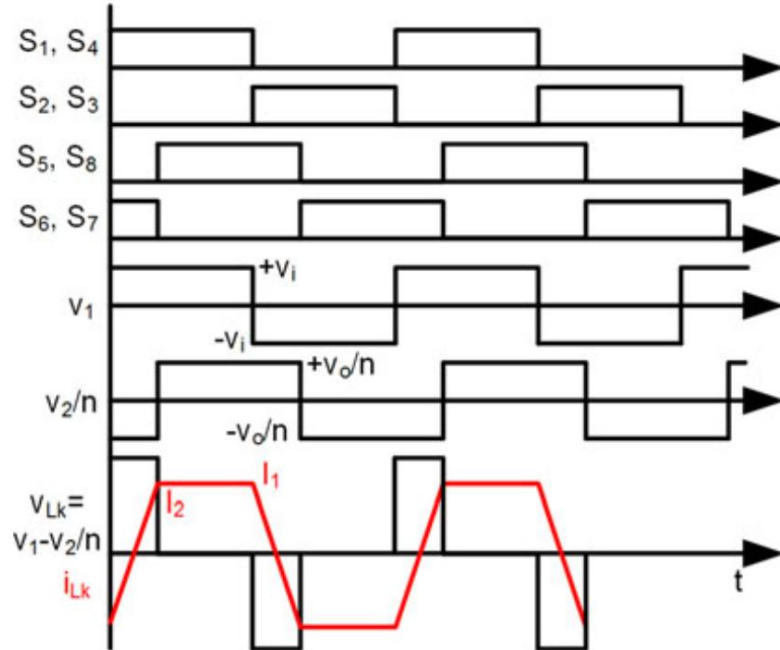


Fig. 3.4. DAB waveforms with SPS in [2]

- Due to the possible combinations of voltage values at the DC ports of both bridges and the turns ratio, the ratio of applied voltages M is defined as follows:

$$M = \frac{V_o}{V_i \cdot n} \quad (3.1)$$

- The values of currents I_1 and I_2 of current i_L are given by:

$$I_1 = \frac{T}{2L_k} \left(2 \frac{V_o}{n} d + V_i - \frac{V_o}{n} \right) = \frac{TV_i}{2L_k} (2Md + 1 - M) \quad (3.2)$$

$$I_2 = \frac{T}{2L_k} \left(2V_i d - V_i + \frac{V_o}{n} \right) = \frac{TV_o}{2L_k} (2d - 1 + M) \quad (3.3)$$

- The average values of the input current i_i to HB1 and output i_o to HB2 are:

$$I_i = \frac{(1-d)dTV_o}{nL_k} \quad (3.4) \quad I_o = \frac{(1-d)dTV_i}{nL_k} \quad (3.5)$$

- Then the average power supplied by HB1 is:

$$P_o = I_o V_o = \frac{V_i V_o T d (1-d)}{nL_k} \quad (3.6)$$

- Applying the base nomenclature ($\varphi/2\pi (b) = d/2$ ([2])), we obtain:

$$P_{SPS} = \frac{n_1}{n_2} \frac{V_1 V_2}{2\pi^2 f_s L} \varphi (\pi - \varphi) \quad (3.7)$$

Section 4: Soft Switching with SPS

Dual Active Bridge



DAB – Soft Switching Techniques with SPS

1. ZVS (Zero Voltage Switching)

- The MOSFET turns on when the voltage at its terminals is practically zero.
- This is achieved by using the energy stored in the transformer's leakage inductance and in the line inductance to charge and discharge the parasitic capacitances of the MOSFETs.
- In the DAB:
 - When there is a phase shift between the bridges, current flows through the transformer.
 - This current allows the voltage at the drain to shift towards 0 V before turning on a MOSFET (because the current charges/discharges the capacitance of the device).
 - Thus, the MOSFET turns on without voltage → ZVS achieved.

2. ZCS (Zero Current Switching)

- The MOSFET turns off when the current flowing through it is zero.
- This is less common in DAB because the topology favors ZVS, but it can occur under low load conditions or in special modes.
- In the DAB:
 - If the phase shift between bridges is small and/or the duty cycles are modified, the current may reach zero at the exact moment of shutdown.
 - This reduces shutdown losses, although it is not always achieved across the entire operating range.

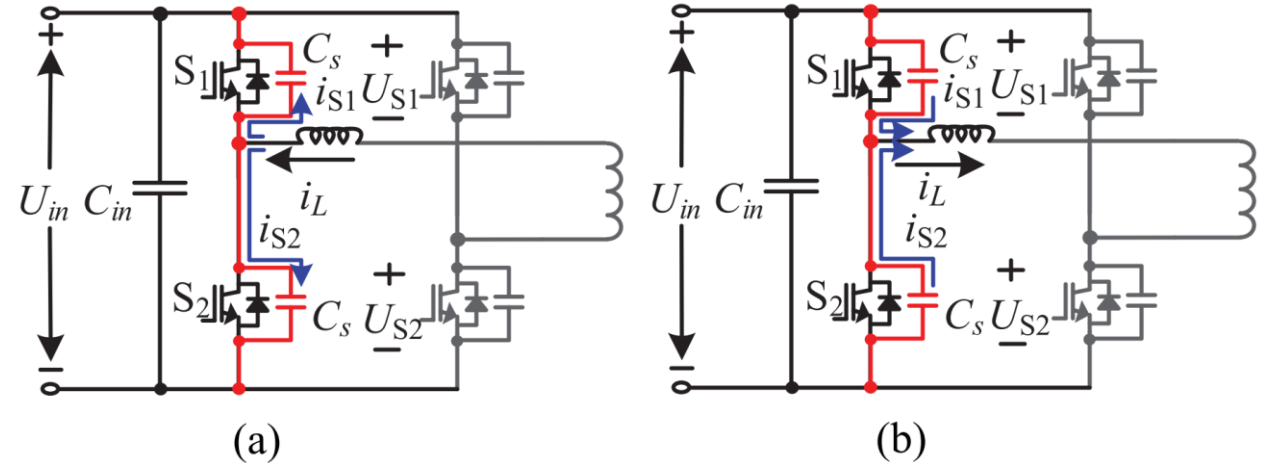


Fig. 4.1. ZVS operation in the non-resonant DAB [5]
(a) ZVS for upper switch (b) ZVS for lower switch

- In order for ZVS to be achieved, not only does current need to flow, but it must also have the minimum value required to transfer the load between the capacitances of the devices:

$$i_{L_min_p} = 2 \sqrt{\frac{U_{in} U_o C_s}{L}} \quad (4.1)$$

$$i_{L_min_s} = 2 \sqrt{\frac{U_{in} U_o C_s}{NL}} \quad (4.2)$$

DAB – Soft Switching Techniques: signals on ZVS

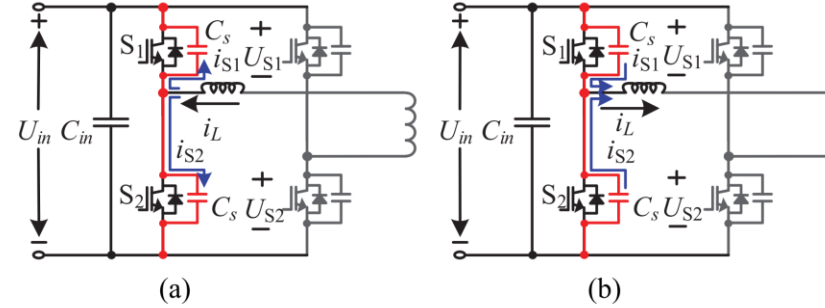


Fig. 17. ZVS performances for the NSDAB dc-dc converter. (a) ZVS for upper switch. (b) ZVS for lower switch.

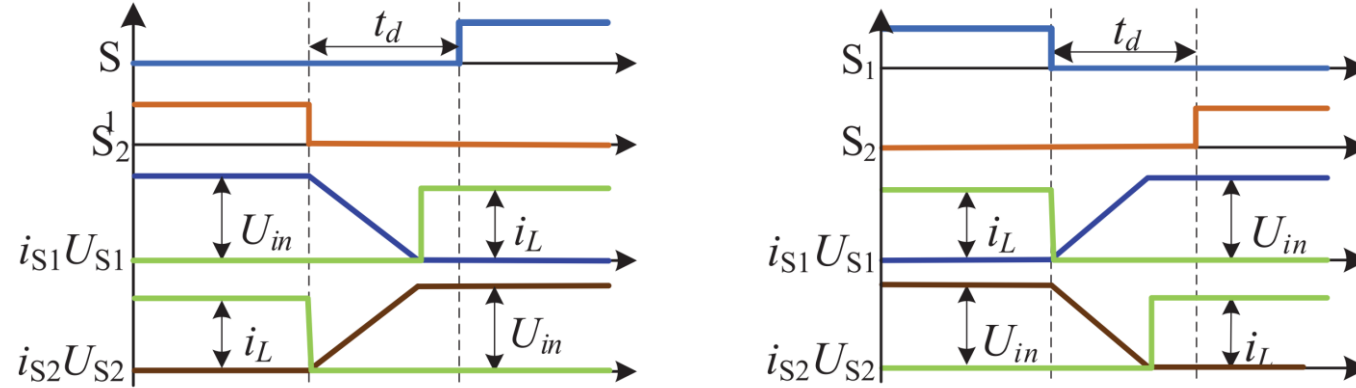


Fig. 18. Working states of S_1 and S_2 during ZVS performance. (a) Signals for upper-switch ZVS. (b) Signals for lower-switch ZVS.

DAB – Conditions for Soft Switching with SPS in the DAB

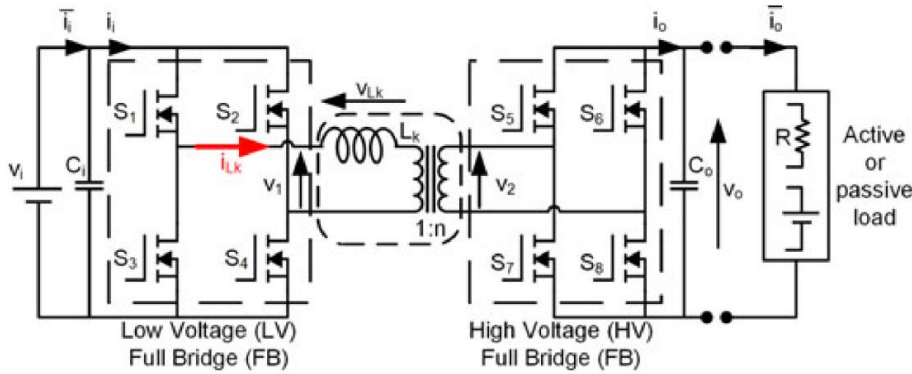


Fig. 4.2. Dual Active Bridge Topology in [2]

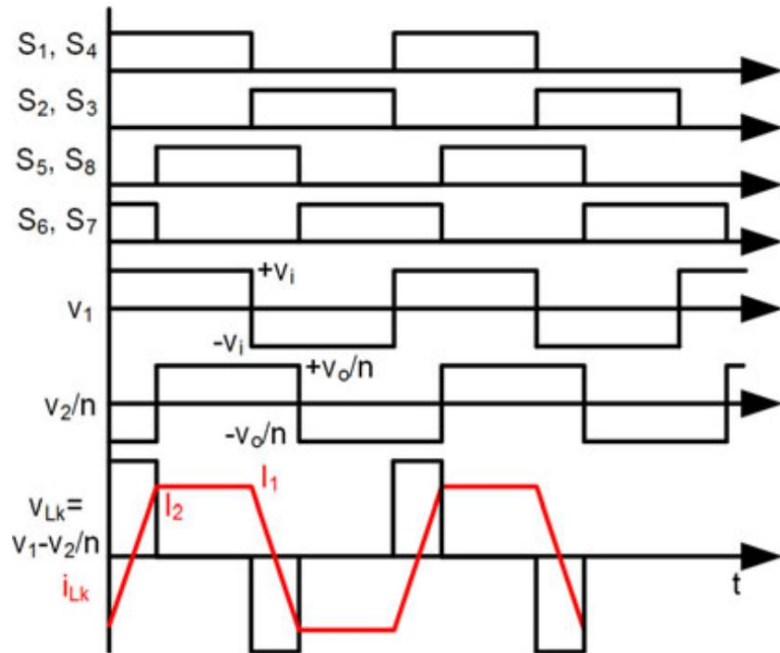


Fig. 4.3. DAB waveforms with SPS in [2]

- The primary semiconductors will turn on with ZVS whenever they turn off with positive current, i.e., $I_1 > 0$, from which (4.3) arises as a necessary condition, assuming that i_L satisfies the charge/discharge condition of (4.1) and (4.2). To achieve ZCS in the primary bridge, $I_1 = 0$ is required, so ZVS can only be achieved in the operating range.
- The secondary semiconductors will turn on with ZVS under the same condition, i.e., it is necessary that $I_2 > 0$, from which (4.4) arises as a necessary condition. For ZCS in the secondary bridge, $I_2 = 0$ is required, so ZVS can only be achieved in the operating range.
- Fig. 4.4 shows the ZVS range in the operating range of the DAB with SPS with different values of the M factor. Obtained according to (3.1), (4.3) and (4.4).

$$M = \frac{V_o}{V_i \cdot n} \quad (3.1)$$

$$I_1 > 0 \leftrightarrow d > \frac{M-1}{2M} \quad (4.3)$$

$$I_2 > 0 \leftrightarrow d > \frac{1-M}{2} \quad (4.4)$$

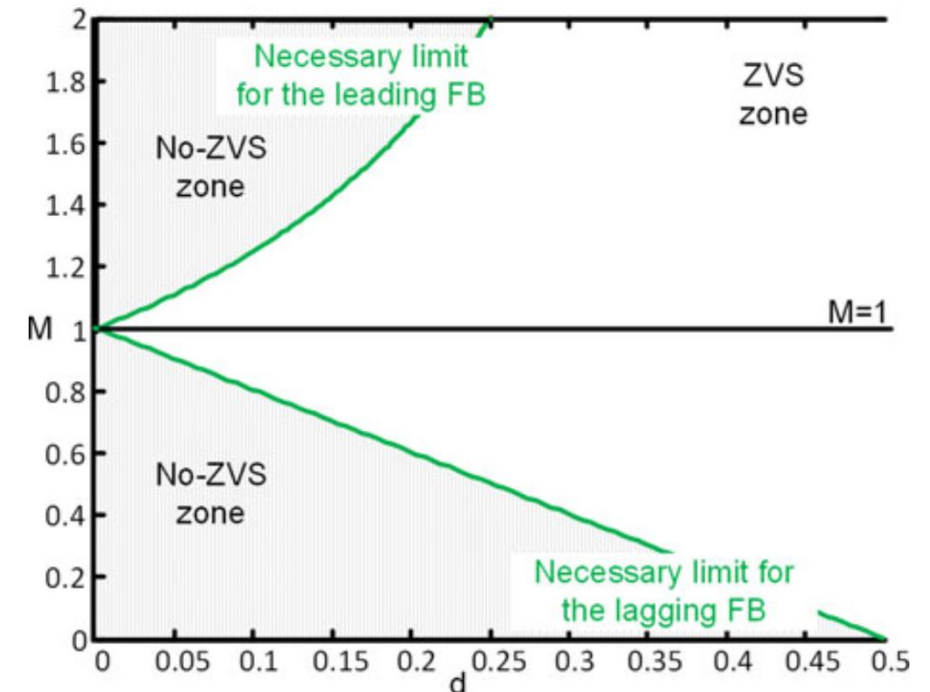


Fig. 4.4. ZVS DAB range representation with SPS[2]

Section 5: Control strategies with SPS

Dual Active Bridge



DAB – SPS Control [2,3]

- [3] shows the DAB with its two connection modes on its secondary port, establishing two control possibilities: current control and voltage control.
- To implement a control strategy in the DAB, it is necessary to understand its dynamic behavior, which according to the modeling is defined by (5.1):

$$P_{SPS} = \frac{n_1}{n_2} \frac{V_i V_o}{2\pi^2 f_s L} \varphi(\pi - \varphi) \quad (5.1)$$

- According to (5.1), the power transferred in the DAB can be plotted for its operating range. Fig. 5.2 shows this dynamic, using a system with a maximum power of 2.5 kW as an example.
- As can be seen, given the parabolic shape of the power dynamics, the system loses linearity in the areas closest to maximum power. Therefore, the literature recommends sizing the DAB to operate at nominal power with a phase angle less than the maximum of 90°, adjusting between 72° and 63° to achieve a better linearity operating zone, which is important for implementing linear control schemes. [2, 3]

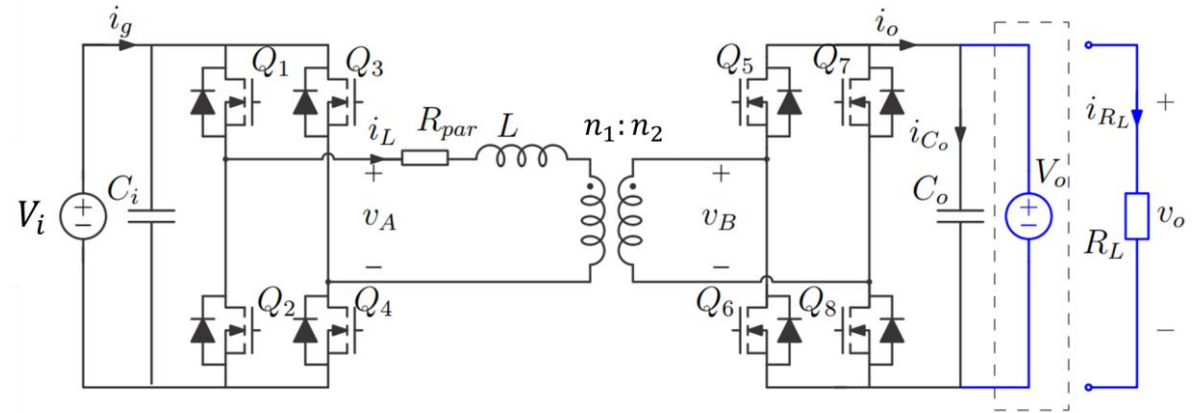


Fig. 5.1. Dual Active Bridge Topology in [3]

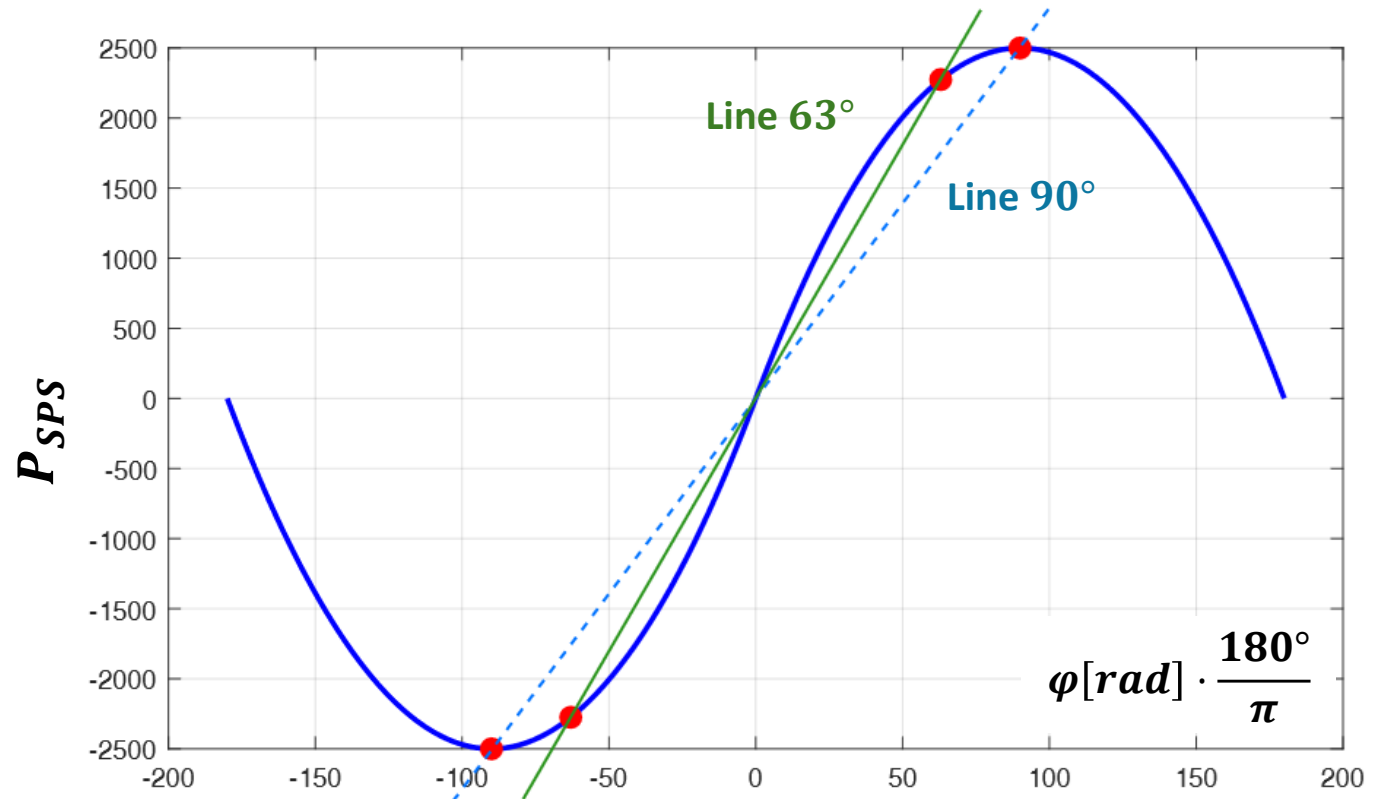


Fig. 5.2. Transfer power in the DAB

DAB – SPS Current Control [2,3]

- From (5.1), the output current can be expressed in (5.2). Likewise, in (5.3) is expressed the current at a fixed operating point ($\varphi = \Phi$).

$$I_o = \frac{P_{SPS}}{V_o} = \frac{n_1}{n_2} \frac{V_i}{2\pi^2 f_s L} \varphi(\pi - \varphi) \quad (5.2)$$

$$I_o^*(\Phi) = \frac{n_1}{n_2} \frac{V_i}{2\pi^2 f_s L} \Phi(\pi - \Phi) \quad (5.3)$$

- It is then possible to obtain analytically in (5.4) the phase angle for a specific reference current (I_o^*):

$$\Phi = \frac{\pi}{2} - \sqrt{\left(\frac{\pi}{2}\right)^2 - \|I_o^*\| \cdot \left(\frac{n_2}{n_1} \frac{2\pi^2 f_s L}{V_i}\right)} \quad (5.4)$$

- In [3], the DAB is modeled in phasor form and the transfer function of (5.5) for the output current ($G_{i_o\varphi}$) is obtained analytically, based on linearization at a stable operating point, which can be calculated/verified with (5.4), considering the recommended angular range for the DAB to operate nominally with $\Phi = 63^\circ \cdot \frac{\pi}{180^\circ}$.

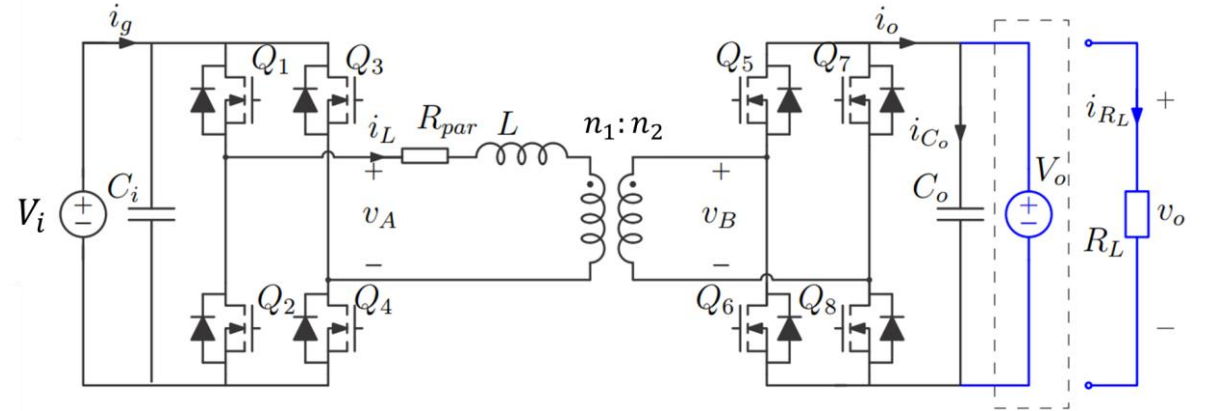


Fig. 5.1. Dual Active Bridge Topology in [3]

$$G_{i_o\varphi}(s) = \frac{\hat{i}_o}{\hat{\varphi}} = V_i \frac{8}{\pi^2} \frac{n_2}{n_1} \frac{-\sin(\Phi) Ls + \omega_s L \cos(\Phi) - R_{par} \sin(\Phi)}{L^2 s^2 + 2R_{par} Ls + R_{par}^2 + \omega_s^2 L^2} \quad (5.5)$$

- A PI controller can then be tuned analytically or using tools such as Sisotool in MATLAB. This allows the control scheme shown in Fig. 5.3 to be formed.

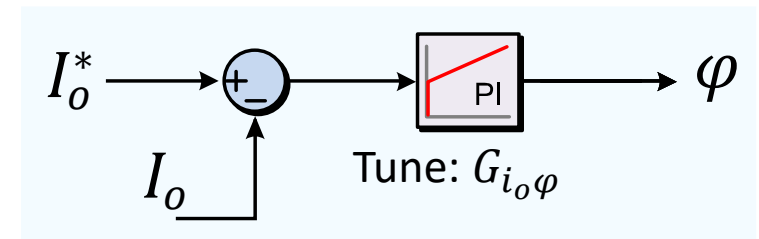


Fig. 5.3. Current control scheme

DAB – SPS Voltage control [2,3]

- Considering now the voltage control of the secondary port capacitor in Fig. 5.1, analyzing the capacitor node yields its transfer function in (5.6).

$$G_{v_o}(s) = \frac{V_o(s)}{I_o(s) - I_{R_L}(s)} = \frac{1}{sC_o} \quad (5.6)$$

- Based on G_{v_o} , it is possible to tune a PI controller for voltage control. Therefore, two control schemes are proposed for DAB:
 - 1) Nested control in Fig. 5.4 with an internal current control loop from the plant $G_{i_o\phi}$.
 - 2) Fig. 5.5 using the analytical expression in (5.7) to find the phase angle.

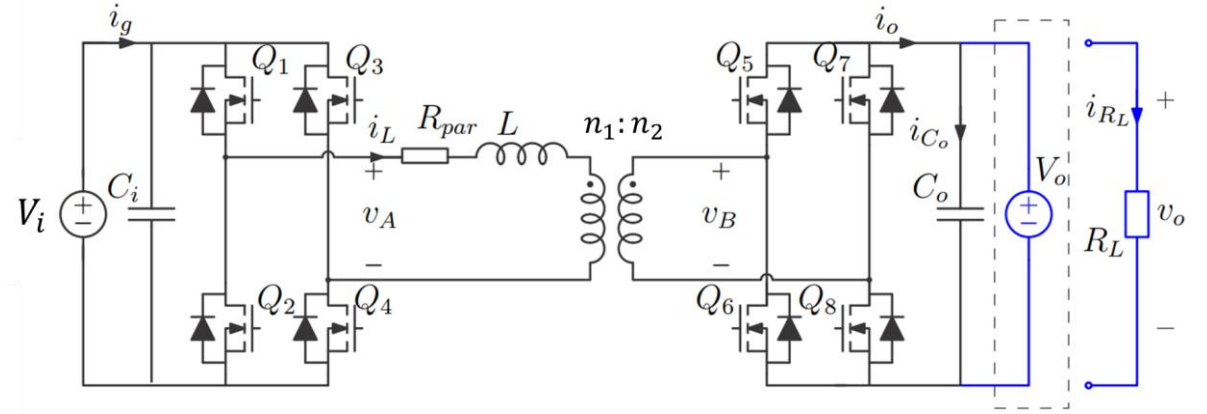


Fig. 5.1. Dual Active Bridge Topology in [3]

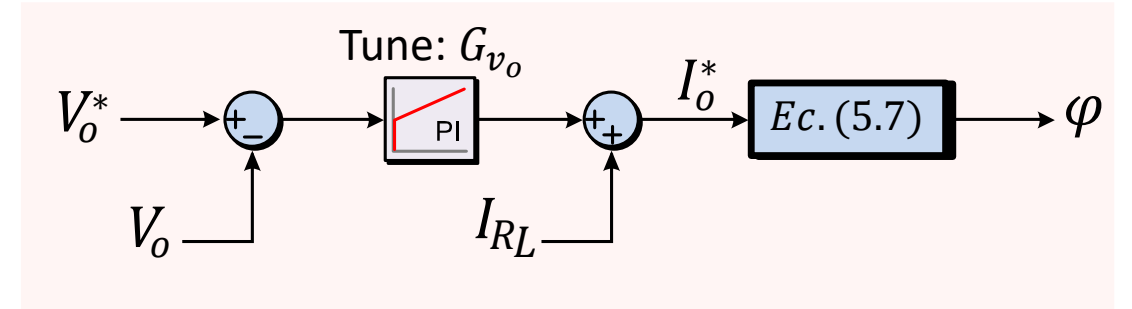


Fig. 5.5. Scheme 2) Voltage control with analytical equation

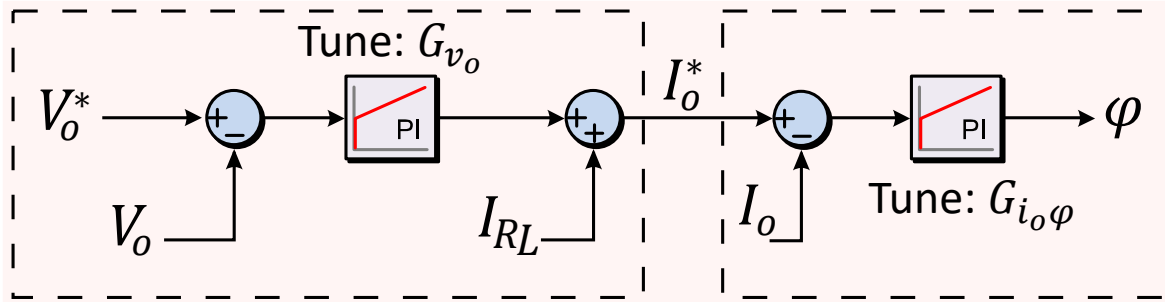


Fig. 5.4. Scheme 1) Nested voltage-current control

$$\begin{aligned} \varphi^* &= \left(\frac{\pi}{2} - \sqrt{\left(\frac{\pi}{2}\right)^2 - \|I_o^*\| \cdot \left(\frac{n_2}{n_1} \frac{2\pi^2 f_s L}{V_i}\right)} \right), \quad \text{for } I_o^* \geq 0 \text{ [A]} \\ \varphi^* &= -\left(\frac{\pi}{2} - \sqrt{\left(\frac{\pi}{2}\right)^2 - \|I_o^*\| \cdot \left(\frac{n_2}{n_1} \frac{2\pi^2 f_s L}{V_i}\right)} \right), \quad \text{for } I_o^* < 0 \text{ [A]} \end{aligned} \quad (5.7)$$

Section 6: Analytical Modeling with TPS

Dual Active Bridge



DAB – TPS modeling in [4]

- [4] presents a study and analytical modeling of power flow and AC current waveform, segmenting the operating range into switching modes.
- The equivalence of the base nomenclature [b] with the variables of the article [4] is:
 - ❖ Turn ratio: $N = n_1/n_2$ [b] = $1/n$ [4]
 - ❖ Duty cycle: D_i [b] = $D_i/2$ [4]
 - ❖ Phase shift: $d_{12} = \varphi/2\pi$ [b] = $\varphi/2$ [4]
(In [4], φ is defined as time phase shift)

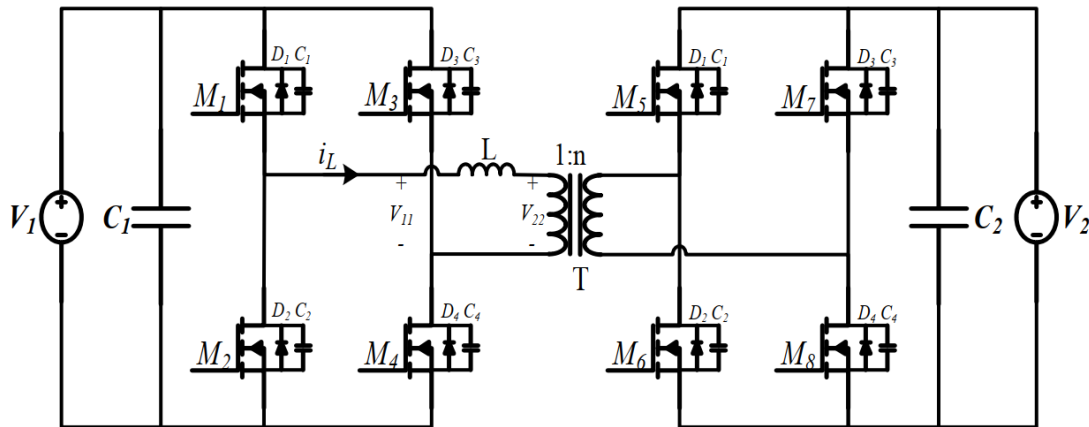


Fig. 6.2. Dual Active Bridge Topology in [4]

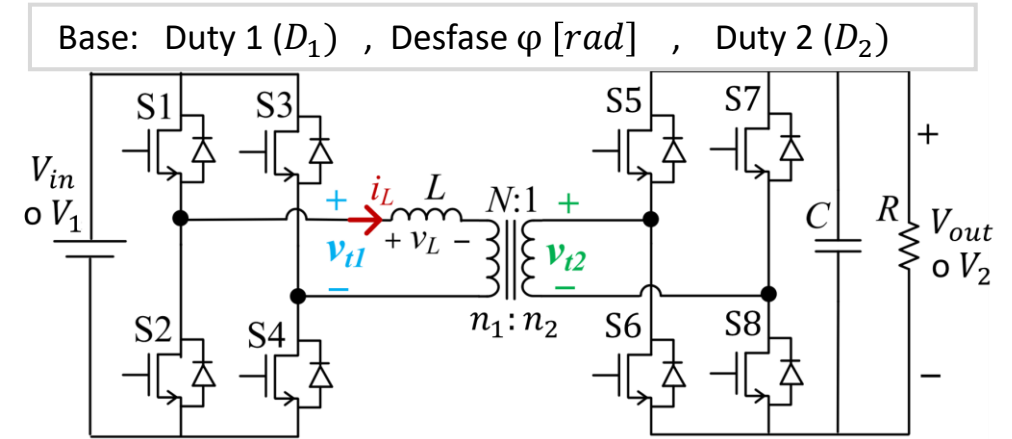


Fig. 6.1. Dual Active Non-Resonant Bridge Topology with base nomenclature [6]

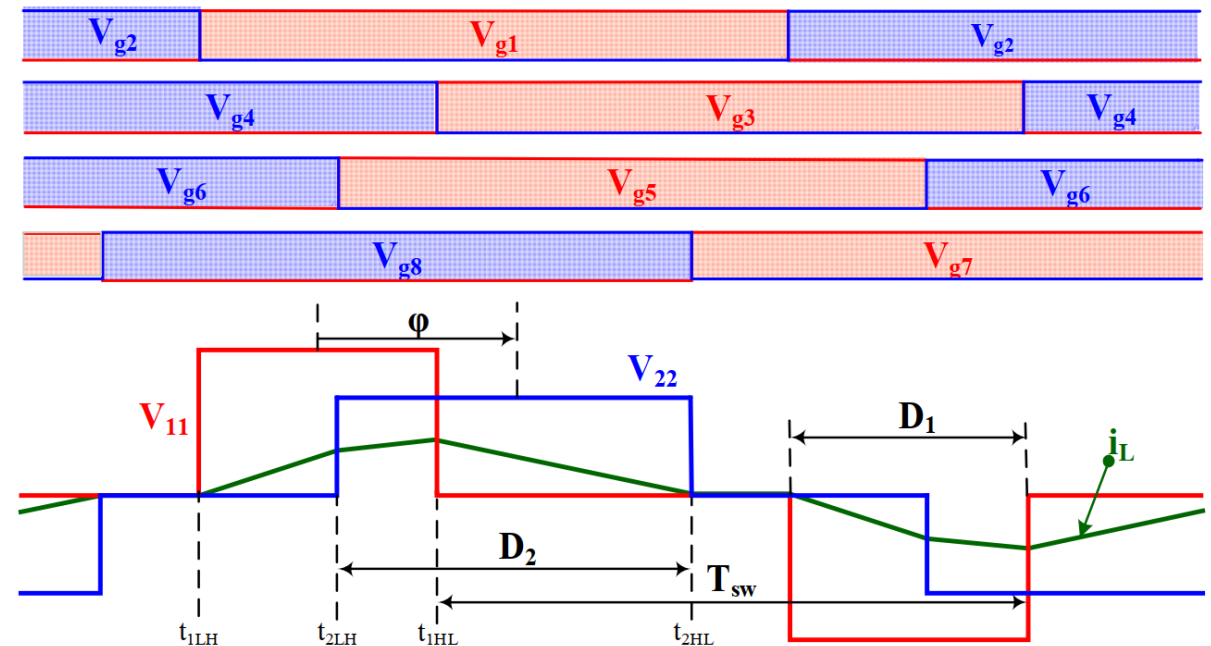


Fig. 6.3. DAB waveforms with TPS in [4]

DAB – TPS modeling in [4]

- The equivalence of the base nomenclature [b] with the variables of the article [4] is:
 - ❖ Turn ratio: $N = n_1/n_2$ [b] = $1/n$ [4]
 - ❖ Duty cycle: D_i [b] = $D_i/2$ [4]
 - ❖ Phase shift: $d_{12} = \varphi/2\pi$ [b] = $\varphi/2$ [4]
- Table 6.1 shows the main switching modes according to [4]. Of these, mode II is taken as the main case for evaluation.

Table 6.1. Operating cases [4]

Cases	Conditions
I	$V_1 \geq V_2/n$ & $D_1 > D_2$
II	$V_1 \geq V_2/n$ & $D_1 \leq D_2$
III	$V_1 < V_2/n$ & $D_1 > D_2$
IV	$V_1 < V_2/n$ & $D_1 \leq D_2$

- Fig. 6.4 shows the voltage waveforms of the AC ports seen from the primary port and the current i_L with the DAB in the case of operation II and for different phase shift values. Table 6.2 defines these operating ranges and specifies the power and currents.

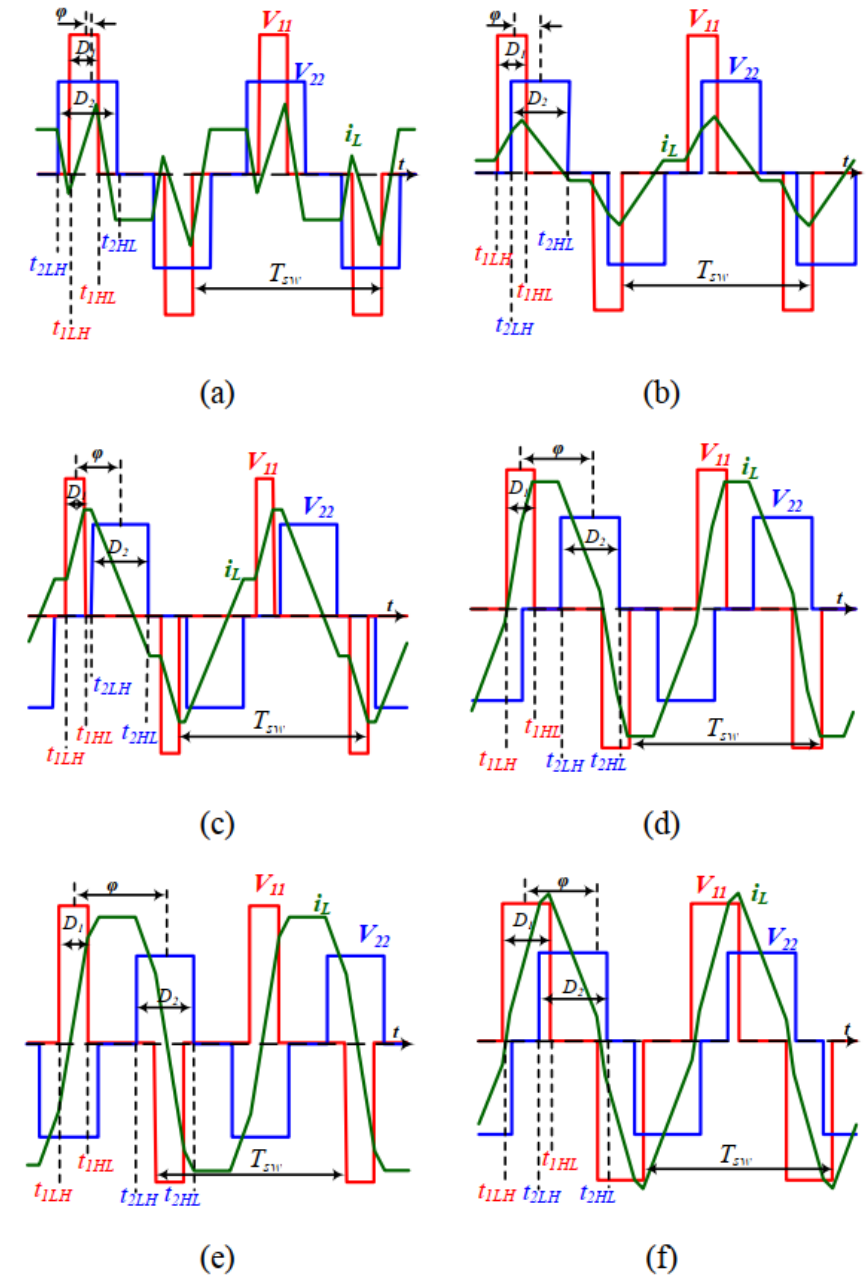


Fig. 6.4. Case II: Switching modes for $\varphi > 0$. (a) φ_1 . (b) φ_2, φ_2^* . (c) φ_3 . (d) φ_4 . (e) φ_5 . (f) φ_3^* . [4]

DAB – TPS modeling in [4]

- The equivalence of the base nomenclature [b] with the variables of the article [4] is:
 - ❖ Turn ratio: $N = n_1/n_2 [b] = \mathbf{1/n [4]}$
 - ❖ Duty cycle: $D_i [b] = \mathbf{D_i/2 [4]}$
 - ❖ Phase shift: $d_{12} = \varphi/2\pi [b] = \mathbf{\varphi/2 [4]}$
- Table 6.1 shows the main switching modes according to [4]. Of these, mode II is taken as the main case for evaluation.

Table 6.1. Operating cases [4]

Cases	Conditions
I	$V_1 \geq V_2/n \ \& \ D_1 > D_2$
II	$V_1 \geq V_2/n \ \& \ D_1 \leq D_2$
III	$V_1 < V_2/n \ \& \ D_1 > D_2$
IV	$V_1 < V_2/n \ \& \ D_1 \leq D_2$

- Fig. 6.4 shows the voltage waveforms of the AC ports seen from the primary port and the current i_L with the DAB in the case of operation II and for different phase shift values. Table 6.2 defines these operating ranges and specifies the power and currents.

Table 6.2. Case II: Range, currents, and powers for switching modes [4]

φ	Range	Current	Power
φ_1	$\left\langle 0; \frac{D_2 - D_1}{2} \right\rangle$	$i_L(t_{1LH}) = -\frac{D_1 \cdot V_1 \cdot n - 2 \cdot V_2 \cdot \varphi - D_1 \cdot V_2}{4 \cdot L \cdot f_{sw} \cdot n}$ $i_L(t_{1HL}) = \frac{D_1 \cdot V_1 \cdot n + 2 \cdot V_2 \cdot \varphi - D_1 \cdot V_2}{4 \cdot L \cdot f_{sw} \cdot n}$ $i_L(t_{2LH}) = -i_L(t_{2HL}) = \frac{D_1 \cdot V_1 \cdot n - D_2 \cdot V_2}{4 \cdot L \cdot f_{sw} \cdot n}$	$\frac{V_1 \cdot V_2 \cdot D_1 \cdot \varphi}{2 \cdot L \cdot f_{sw} \cdot n}$
φ_2	$\left\langle \frac{D_2 - D_1}{2}; \frac{D_2 + D_1}{2} \right\rangle$	$i_L(t_{1LH}) = -i_L(t_{2HL}) = -\frac{D_1 \cdot V_1 \cdot n - D_2 \cdot V_2}{4 \cdot L \cdot f_{sw} \cdot n}$ $i_L(t_{1HL}) = \frac{D_1 \cdot V_1 \cdot n + 2 \cdot V_2 \cdot \varphi - D_1 \cdot V_2}{4 \cdot L \cdot f_{sw} \cdot n}$	$\frac{V_1 \cdot V_2}{4 \cdot L \cdot f_{sw} \cdot n} \left[\varphi \cdot (D_1 + D_2 - \varphi) - \frac{(D_1 - D_2)^2}{4} \right]$
φ_2^*	$\left\langle \frac{D_2 - D_1}{2}; 1 - \frac{D_2 + D_1}{2} \right\rangle$	$i_L(t_{2LH}) = -\frac{D_2 \cdot V_1 \cdot n - 2 \cdot V_2 \cdot \varphi - D_2 \cdot V_2}{4 \cdot L \cdot f_{sw} \cdot n}$	
φ_3	$\left\langle \frac{D_2 + D_1}{2}; 1 - \frac{D_2 + D_1}{2} \right\rangle$	$i_L(t_{1LH}) = -i_L(t_{2HL}) = -\frac{D_1 \cdot V_1 \cdot n - D_2 \cdot V_2}{4 \cdot L \cdot f_{sw} \cdot n}$ $i_L(t_{1HL}) = i_L(t_{2LH}) = \frac{D_1 \cdot V_1 \cdot n + D_2 \cdot V_2}{4 \cdot L \cdot f_{sw} \cdot n}$	$\frac{V_1 \cdot V_2 \cdot D_1 \cdot D_2}{4 \cdot L \cdot f_{sw} \cdot n}$
φ_3^*	$\left\langle 1 - \frac{D_2 + D_1}{2}; \frac{D_2 + D_1}{2} \right\rangle$	$i_L(t_{1LH}) = -\frac{D_1 \cdot V_1 \cdot n + D_1 \cdot V_2 - 2 \cdot V_2 \cdot (1 - \varphi)}{4 \cdot L \cdot f_{sw} \cdot n}$ $i_L(t_{1HL}) = \frac{D_1 \cdot V_1 \cdot n - D_1 \cdot V_2 + 2 \cdot V_2 \cdot \varphi}{4 \cdot L \cdot f_{sw} \cdot n}$ $i_L(t_{2LH}) = -\frac{D_2 \cdot V_1 \cdot n - D_2 \cdot V_2 - 2 \cdot V_1 \cdot n \cdot \varphi}{4 \cdot L \cdot f_{sw} \cdot n}$ $i_L(t_{2HL} - \frac{T_{sw}}{2}) = \frac{D_2 \cdot V_1 \cdot n + D_2 \cdot V_2 - 2 \cdot V_1 \cdot n \cdot (1 - \varphi)}{4 \cdot L \cdot f_{sw} \cdot n}$	$\frac{V_1 \cdot V_2}{2 \cdot L \cdot f_{sw} \cdot n} \left[\varphi \cdot (1 - \varphi) - \frac{(D_1 - 1)^2 + (D_2 - 1)^2}{4} \right]$
φ_4	$\left\langle 1 - \frac{D_2 + D_1}{2}; 1 - \frac{D_2 - D_1}{2} \right\rangle$	$i_L(t_{1LH}) = -\frac{D_1 \cdot V_1 \cdot n + D_1 \cdot V_2 - 2 \cdot V_2 \cdot (1 - \varphi)}{4 \cdot L \cdot f_{sw} \cdot n}$ $i_L(t_{1HL}) = i_L(t_{2LH}) = \frac{D_1 \cdot V_1 \cdot n + D_2 \cdot V_2}{4 \cdot L \cdot f_{sw} \cdot n}$ $i_L(t_{2HL} - \frac{T_{sw}}{2}) = \frac{D_2 \cdot V_1 \cdot n + D_2 \cdot V_2 - 2 \cdot V_1 \cdot n \cdot (1 - \varphi)}{4 \cdot L \cdot f_{sw} \cdot n}$	$\frac{V_1 \cdot V_2}{4 \cdot L \cdot f_{sw} \cdot n} \left[(1 - \varphi) \cdot (D_1 + D_2 - \varphi) - \frac{(D_1 - D_2)^2}{4} \right]$
φ_5	$\left\langle 1 - \frac{D_2 - D_1}{2}; 1 \right\rangle$	$i_L(t_{1LH}) = -\frac{D_1 \cdot V_1 \cdot n + D_1 \cdot V_2 - 2 \cdot V_2 \cdot (1 - \varphi)}{4 \cdot L \cdot f_{sw} \cdot n}$ $i_L(t_{1HL}) = \frac{D_1 \cdot V_1 \cdot n + D_1 \cdot V_2 + 2 \cdot V_2 \cdot (1 - \varphi)}{4 \cdot L \cdot f_{sw} \cdot n}$ $i_L(t_{2LH} - \frac{T_{sw}}{2}) = -i_L(t_{2HL} - \frac{T_{sw}}{2}) = -\frac{D_1 \cdot V_1 \cdot n + D_2 \cdot V_2}{4 \cdot L \cdot f_{sw} \cdot n}$	$\frac{V_1 \cdot V_2 \cdot D_2 \cdot (1 - \varphi)}{2 \cdot L \cdot f_{sw} \cdot n}$

DAB – TPS modeling in [5]

- [5] presents an overview of the most relevant DAB modulations and control schemes. It analyzes the power flow and current waveform, identifying the multiple possible switching modes.
- The equivalence of the base nomenclature [b] with the variables of the article [5] is:
 - ❖ Turn ratio : $N = n_1/n_2$ [b] = $1/n$ [5]
 - ❖ Duty cycle 1 : D_1 [b] = $0.5 - D_1/2$ [5]
 - ❖ Duty cycle 2 : D_2 [b] = $0.5 - (\frac{D_3}{2} - \frac{D_2}{2})$ [5]
 - ❖ Phase shift : d_{12} [b] = $\frac{-D_1 + D_2 + D_3}{4}$ [5]

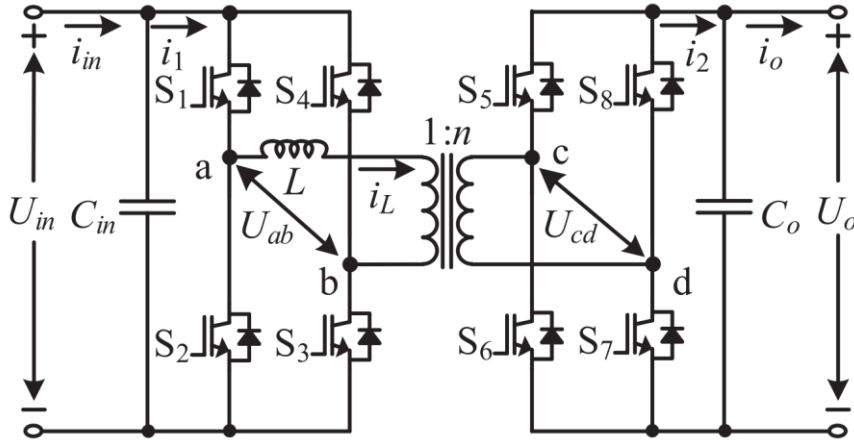


Fig. 6.5. Dual Active Bridge Topology in [5]

Base: Duty 1 (D_1) , Desfase φ [rad] , Duty 2 (D_2)

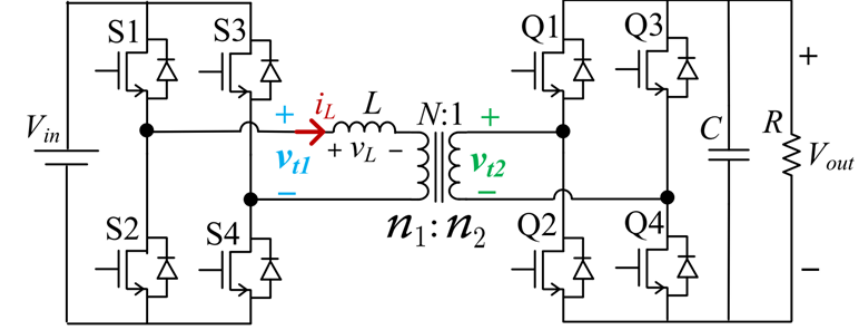


Fig. 6.1. Dual Active Non-Resonant Bridge Topology with base nomenclature [6]

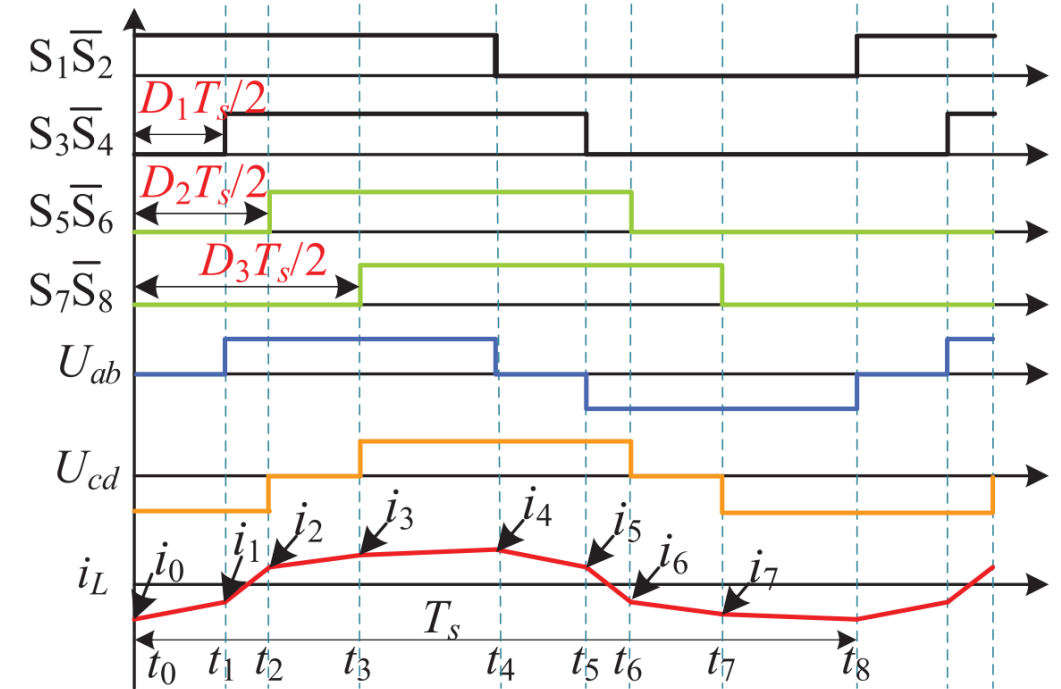


Fig. 6.6. DAB waveforms with TPS in [5]

DAB – TPS modeling in [5]

- The equivalence of the base nomenclature [b] with the variables of the article [5] is:

- ❖ Turn ratio : $N = n_1/n_2 [b] = \mathbf{1/n [5]}$
- ❖ Duty cycle 1 : $D_1 [b] = \mathbf{0.5 - D_1/2 [5]}$
- ❖ Duty cycle 2 : $D_2 [b] = \mathbf{0.5 - (\frac{D_3}{2} - \frac{D_2}{2}) [5]}$
- ❖ Phase shift : $d_{12} [b] = \mathbf{\frac{-D_1 + D_2 + D_3}{4} [5]}$

- Fig. 12 from [5], presented in this slide, illustrates the 18 non-redundant switching modes that can be formed in the DAB with TPS modulation. The different current waveforms that can be generated according to the combinations that arise from varying the duty cycles and phase shift between the ports can be visualized.

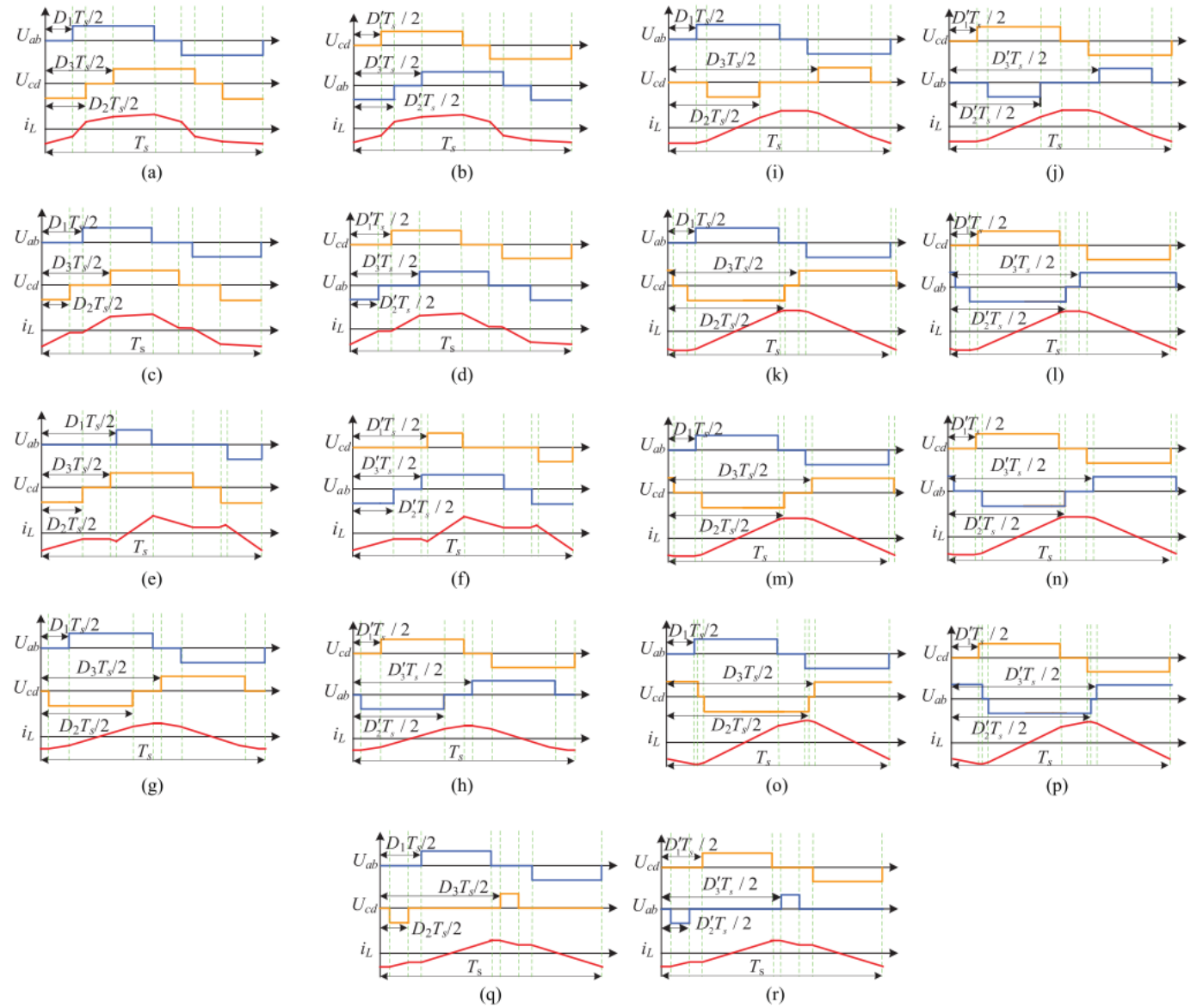


Fig. 12. 18 basic modulation methods for the NSDAB dc-dc converter. (a) $0 \leq D_1 \leq D_2 \leq D_3 \leq 1$. (b) $0 \leq D'_1 \leq D'_2 \leq D'_3 \leq 1$. (c) $0 \leq D_2 \leq D_1 \leq D_3 \leq 1$. (d) $0 \leq D'_2 \leq D'_1 \leq D'_3 \leq 1$. (e) $0 \leq D_2 \leq D_3 \leq D_1 \leq 1$. (f) $0 \leq D'_2 \leq D'_3 \leq D'_1 \leq 1$. (g) $0 \leq D_1 \leq D_2 \leq 1 \leq D_3 \leq 1 + D_1 \leq 2$. (h) $0 \leq D'_1 \leq D'_2 \leq 1 \leq D'_3 \leq 1 + D'_1 \leq 2$. (i) $0 \leq D_1 \leq D_2 \leq 1 \leq 1 + D_1 \leq D_3 \leq 2$. (j) $0 \leq D'_1 \leq D'_2 \leq 1 \leq 1 + D'_1 \leq D'_3 \leq 2$. (k) $0 \leq D_1 \leq 1 \leq D_2 \leq D_3 \leq 1 + D_1 \leq 2$. (l) $0 \leq D'_1 \leq 1 \leq D'_2 \leq D'_3 \leq 1 + D'_1 \leq 2$. (m) $0 \leq D_1 \leq 1 \leq D_2 \leq 1 + D_1 \leq D_3 \leq 2$. (n) $0 \leq D'_1 \leq 1 \leq D'_2 \leq 1 + D'_1 \leq D'_3 \leq 2$. (o) $0 \leq D_1 \leq 1 \leq 1 + D_1 \leq D_2 \leq D_3 \leq 2$. (p) $0 \leq D'_1 \leq 1 \leq 1 + D'_1 \leq D'_2 \leq D'_3 \leq 2$. (q) $0 \leq D_2 \leq D_1 \leq 1 \leq D_3 \leq 1 + D_2 \leq 2$. (r) $0 \leq D'_2 \leq D'_1 \leq 1 \leq D'_3 \leq 1 + D'_2 \leq 2$.

Section 7: Soft Switching with TPS

Dual Active Bridge



DAB – Conditions for **Soft Switching** with TPS in DAB

- Previously, slide 19 explained the ZVS and ZCS soft switching techniques in the DAB. In subsequent analyses presented in various articles, it is assumed that the minimum current value of (4.1) is met for ZVS conditions.
- The analysis in [4], in addition to [5] and [7], will be used as a basis for evaluating soft switching conditions. It indicates that under certain conditions, soft switching is achievable across the entire power range of the DAB.
- Fig. 7.2 shows the voltage and current waveforms i_L that can be generated in the DAB operating with TPS under an RMS current minimization scheme or strategy [5, 6].

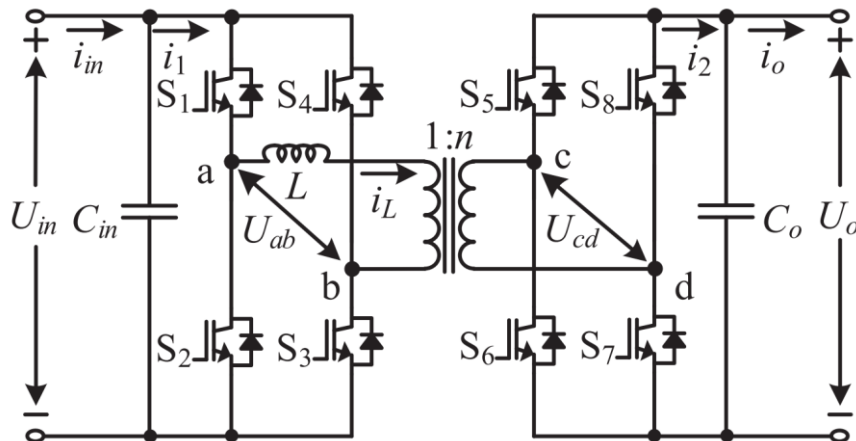
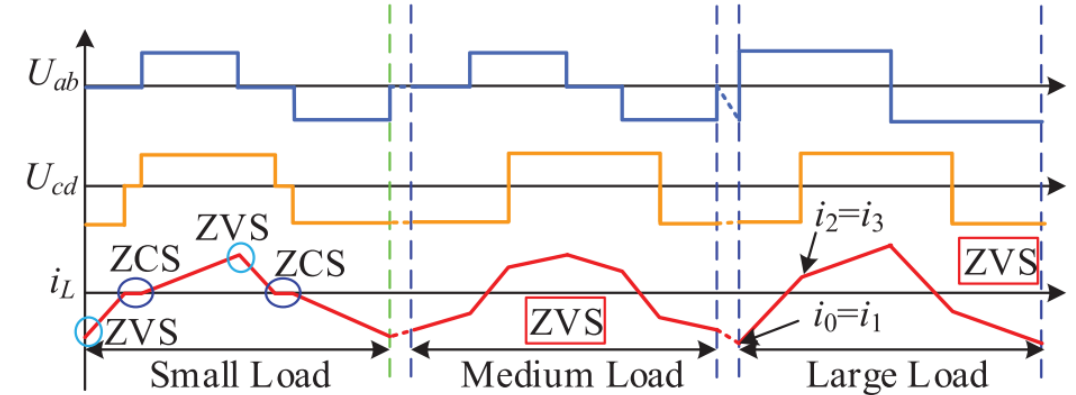
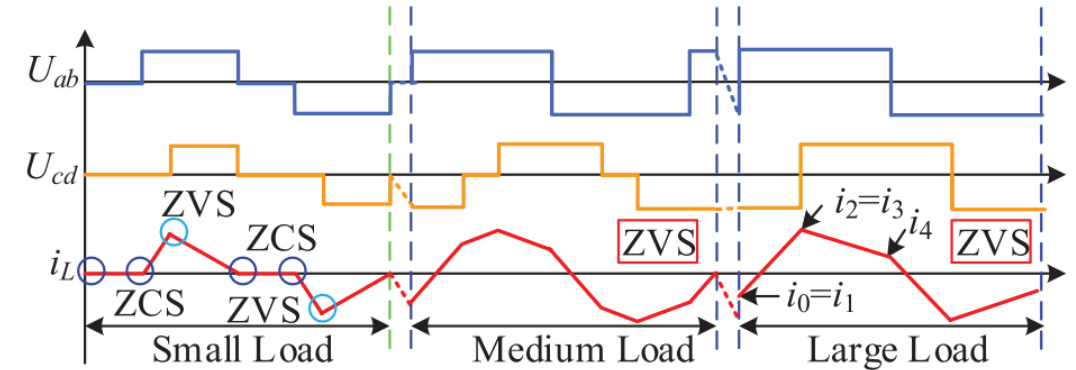


Fig. 7.1. Dual Active Bridge Topology in [5]



(a)



(b)

Fig. 7.2. Triple Phase-shift modulation in “Minimization RMS Current” scheme under different load condition. (a) $k > 1$. (b) $k < 1$. [5]

where $k = \frac{n \cdot U_{in}}{U_o}$, equivalent to $k = 1/M$ with $M = \frac{V_o}{n \cdot V_i}$

DAB – Conditions for **Soft Switching** with TPS in DAB

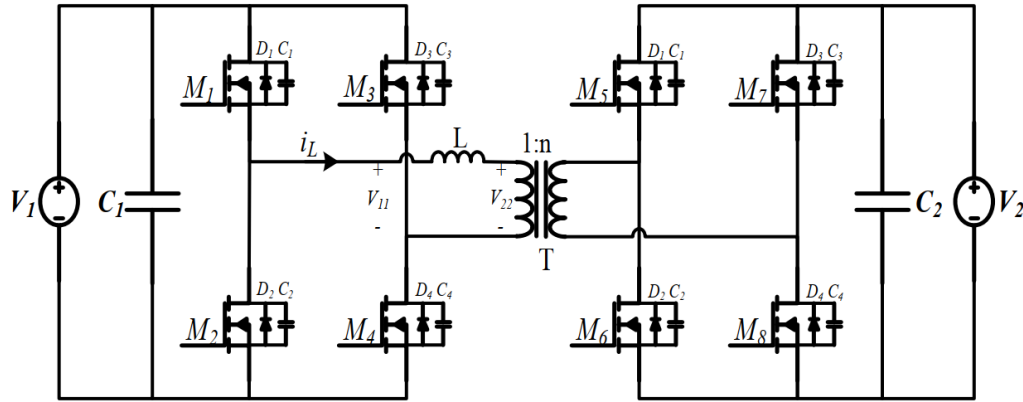


Fig. 7.3. Dual Active Bridge Topology in [4]

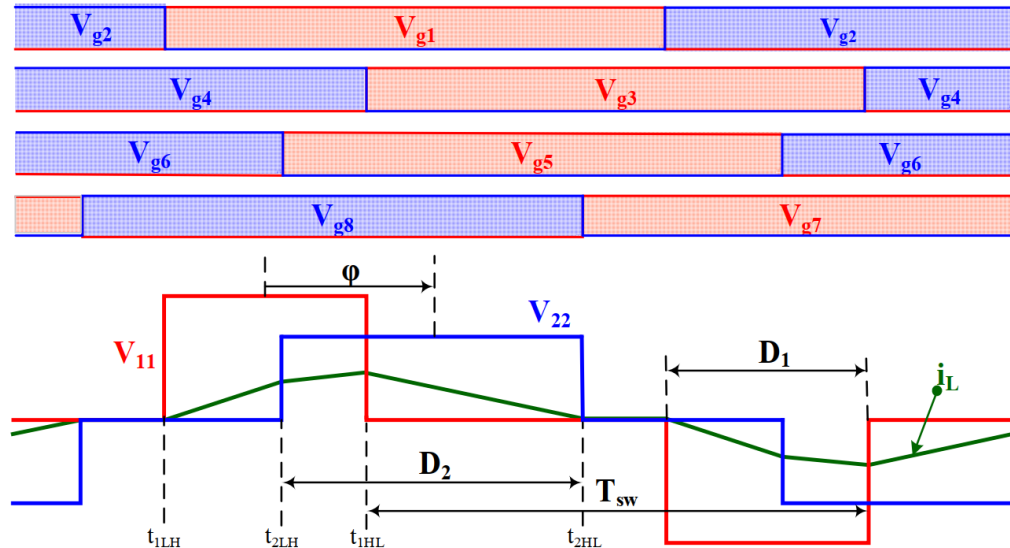


Fig. 7.4. DAB waveforms with TPS in [4]

- Table 7.1 shows the main switching modes according to [4], where mode II is the main evaluation mode.

Table 7.1. Operating cases [4]

Cases	Conditions
I	$V_1 \geq V_2/n$ & $D_1 > D_2$
II	$V_1 \geq V_2/n$ & $D_1 \leq D_2$
III	$V_1 < V_2/n$ & $D_1 > D_2$
IV	$V_1 < V_2/n$ & $D_1 \leq D_2$

- In [4], it is demonstrated that to achieve soft switching conditions in the DAB operating in case II and case III (equivalent to swapping voltage levels), it is sufficient to satisfy relation (7.1). This achieves soft switching across the entire operating range of the DAB.

$$D_1 \cdot V_1 = D_2 \cdot V_2/n \quad (7.1)$$

- In [7], operating zones equivalent to the cases in Table 7.1 in [4] are also presented, and the conditions for ZVS are analyzed. In this study, relation (7.1) is also a solution.

DAB – Conditions for **Soft Switching** with TPS in DAB

- The equivalence of the base nomenclature [b] with the variables of the article [4] is:
 - ❖ Turn ratio: $N = n_1/n_2$ [b] = **$1/n$** [4]
 - ❖ Duty cycle: D_i [b] = **$D_i/2$** [4]
 - ❖ Phase shift: $d_{12} = \varphi/2\pi$ [b] = **$\varphi/2$** [4]

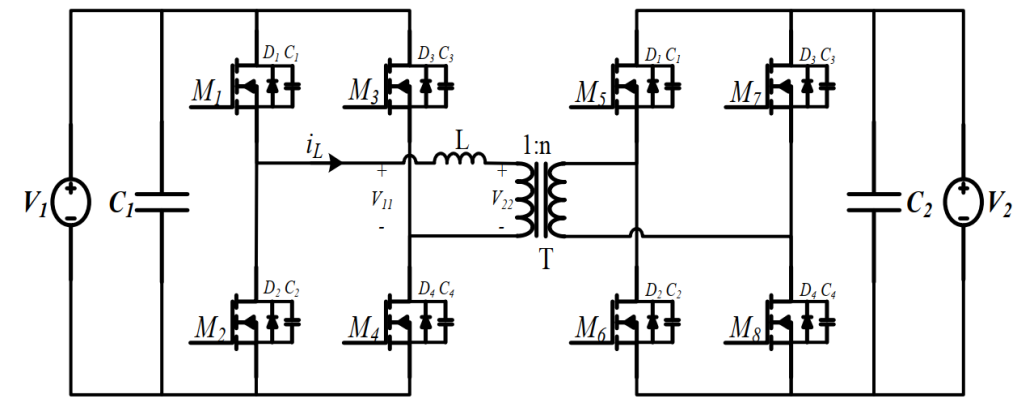


Fig. 7.3. Dual Active Bridge Topology in [4]

Table 7.2. Range, Power, and Switching Type for Cases II and III [4]

φ	Range	Power		Type of switching											
		Case II	Case III	M ₁	M ₂	M ₃	M ₄	M ₅	M ₆	M ₇	M ₈				
φ_1	$\left\langle 0; \frac{ D_2 - D_1 }{2} \right\rangle$	$\frac{V_1^2 \cdot D_1^2 \cdot \varphi}{2 \cdot L \cdot f_{sw} \cdot D_2}$	$\frac{V_1^2 \cdot D_1 \cdot \varphi}{2 \cdot L \cdot f_{sw}}$	ZVS		ZVS		ZCS							
	$\frac{ D_2 - D_1 }{2}$														
φ_2	$\left\langle \frac{ D_2 - D_1 }{2}; \frac{D_2 + D_1}{2} \right\rangle$	$\frac{V_1^2 \cdot D_1}{4 \cdot L \cdot f_{sw} \cdot D_2} \left[\varphi \cdot (D_1 + D_2 - \varphi) - \frac{(D_1 - D_2)^2}{4} \right]$		ZCS								ZCS			
φ_2^*	$\left\langle \frac{ D_2 - D_1 }{2}; 1 - \frac{D_2 + D_1}{2} \right\rangle$														
φ_3	$\left\langle \frac{D_1 + D_2}{2}; 1 - \frac{D_1 + D_2}{2} \right\rangle$	$\frac{V_1^2 \cdot D_1^2}{4 \cdot L \cdot f_{sw}}$						ZVS							
φ_3^*	$\left\langle 1 - \frac{D_1 + D_2}{2}; \frac{D_1 + D_2}{2} \right\rangle$	$\frac{V_1^2 \cdot D_1}{2 \cdot L \cdot f_{sw} \cdot D_2} \left[\varphi \cdot (1 - \varphi) - \frac{(D_1 - 1)^2 + (D_2 - 1)^2}{4} \right]$		ZVS								ZVS			
φ_4	$\left\langle 1 - \frac{D_1 + D_2}{2}; 1 - \frac{ D_1 - D_2 }{2} \right\rangle$	$\frac{V_1^2 \cdot D_1}{4 \cdot L \cdot f_{sw} \cdot D_2} \left[(1 - \varphi) \cdot (D_1 + D_2 + \varphi - 1) - \frac{(D_1 - D_2)^2}{4} \right]$													
φ_5	$\left\langle 1 - \frac{ D_1 - D_2 }{2}; 1 \right\rangle$	$\frac{V_1^2 \cdot D_1 \cdot (1 - \varphi)}{2 \cdot L \cdot f_{sw}}$	$\frac{V_1^2 \cdot D_1^2 \cdot (1 - \varphi)}{2 \cdot L \cdot f_{sw} \cdot D_2}$												

- Table 7.2 shows the type of soft switching achieved by the DAB with TPS in cases II and III, applying the ratio (7.1) to these cases.

$$D_1 \cdot V_1 = D_2 \cdot V_2/n \quad (7.1)$$

Sección 8: General Harmonic Model with TPS

Dual Active Bridge



DAB – Modelamiento TPS en [6]

General Harmonic Model

- The literature also presents a type of modeling based on Fourier theory, expressing voltage waveforms as periodic sinusoidal functions. Although there are various articles and further studies, [6] has been selected.
- The basic nomenclature is that of [6]:
 - ❖ Turn ratio: $N = n_1/n_2$ [b]
 - ❖ Duty cycle: D_i [b]
 - ❖ Phase shift: $d_{12} = \varphi/2\pi$ [b]
- Using Fourier series, AC port voltages are represented by the sum of harmonic components:

$$v_1(t) = V_{dc1} \sum_{k=1,3,5\dots}^{2h+1} \frac{4}{k \cdot \pi} \sin(k\pi D_1) \cos(k(\omega t + \varphi)) \quad (8.1)$$

$$v_2(t) = V_{dc2} \sum_{k=1,3,5\dots}^{2h+1} \frac{4}{k \cdot \pi} \sin(k\pi D_2) \cos(k\omega t) \quad (8.2)$$

Where h is the number of non-zero harmonic components due to the symmetry of the signals.

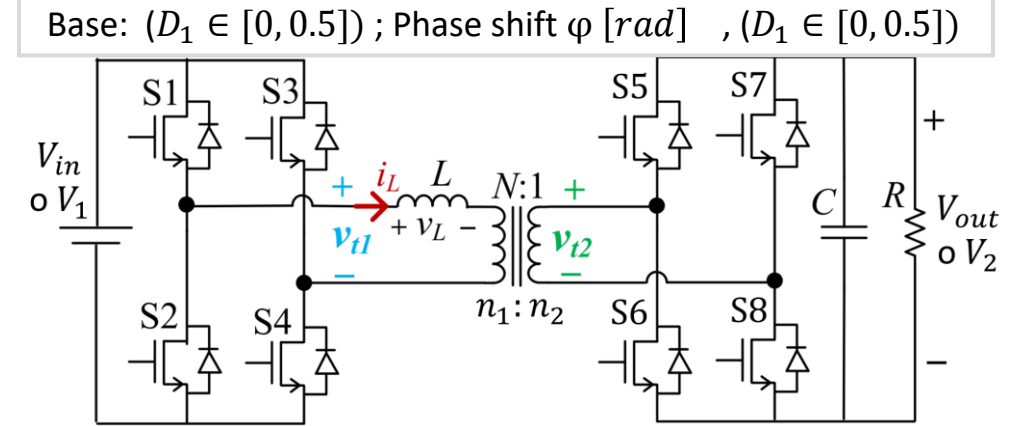


Fig. 8.1. Dual Active Non-Resonant Bridge Topology with base nomenclature [6]

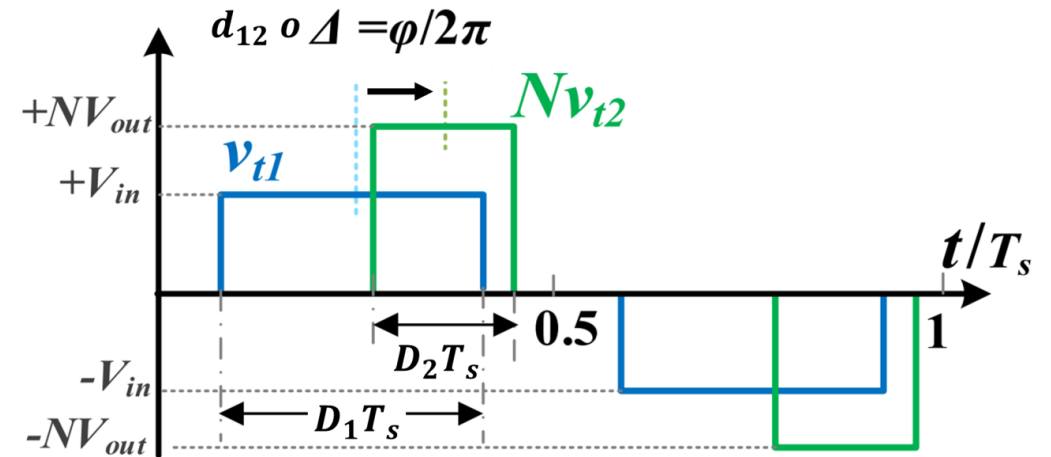


Fig. 8.2. Waveform of voltages at AC ports with TPS [6]

DAB – Modelamiento TPS en [6]

General Harmonic Model

- Using (8.3), the current i_L or i_{L12} can be calculated. Its expression is obtained in (8.4).

$$i_{L_{ij}} = \frac{1}{L_{ij}} \int v_{L_{ij}} dt = \frac{1}{L_{ij}} \int (v_i - Nv_j) dt \quad (8.3)$$

$$i_{L12} = \frac{4}{\pi \omega L_{12}} \sum_{k=1,3,5\dots}^{2h+1} \left[\frac{V_1}{k^2} \sin(k\pi D_1) \sin(k(\omega t + \varphi)) - \frac{NV_2}{k^2} \sin(k\pi D_2) \sin(k\omega t) \right] \quad (8.4)$$

- Then, using (8.5), the average power transferred from the primary port to the secondary port can be calculated, obtaining (8.6).

$$P_{12} = \frac{1}{T} \int_0^T p_{12} dt = \frac{1}{T} \int_0^T (v_1 \cdot i_{12}) dt \quad (8.5)$$

$$P_{12} = \frac{4 \cdot N \cdot V_1 \cdot V_2}{\pi^3 \cdot f_{sw} \cdot L_{12}} \sum_{k=1,3,5\dots}^{2h+1} \frac{1}{k^3} \sin(k\pi D_1) \sin(k\pi D_2) \sin(k\varphi) \quad (8.6)$$

- The more harmonic components are considered in the optimization problem, the better the result obtained for the established average power [6].

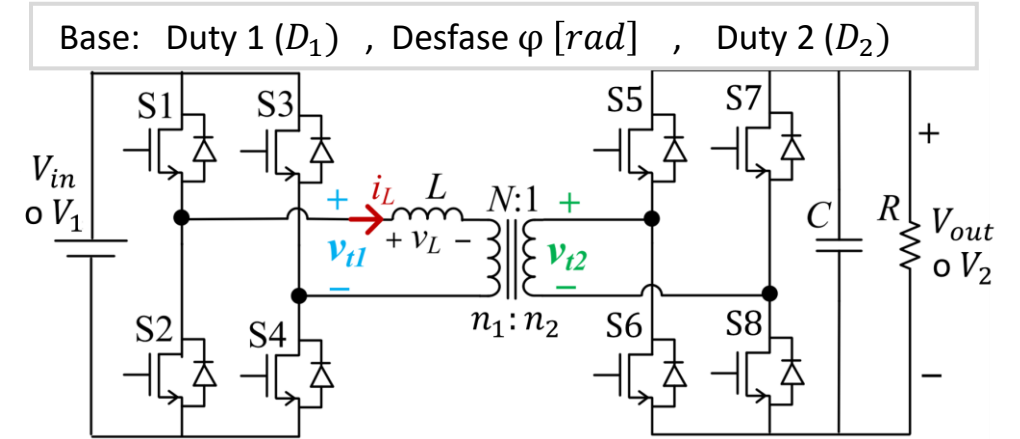


Fig. 8.1. Dual Active Non-Resonant Bridge Topology with base nomenclature [6]

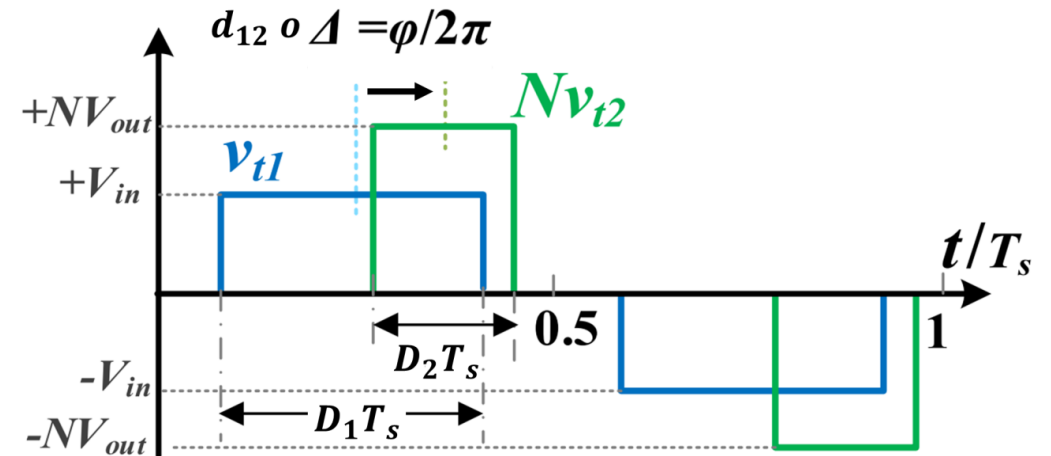


Fig. 8.2. Waveform of voltages at AC ports with TPS [6]

Section 9: Control strategies with TPS

Dual Active Bridge



DAB – Optimization

- To minimize conduction losses, a good approach is to minimize the RMS current squared in the system [6, 7]. The objective function of an optimization can then be defined in (9.1):

$$F_{OBJ}(\mathbf{x}) = I_{L12_{RMS}}^2 = \frac{1}{T} \int_0^T (i_{L12})^2 dt \quad (9.1)$$

- Then, in (9.2), the optimization problem to be solved is posed, which consists of minimizing the RMS current squared using the required transfer power as a constraint, thereby obtaining the values of the vector of variables $\mathbf{x}$ corresponding to the TPS degrees of freedom variables. The problem can be solved numerically with the MATLAB *fmincon* package [6, 7].

$$\text{Optimize} \begin{cases} \min_{\mathbf{x}} F_{OBJ}(\mathbf{x}) \\ \text{s. t.} \\ P(\mathbf{x}) = P_{ref} \\ L_b \leq \mathbf{x} \leq U_b \end{cases} \quad (9.2)$$

- The TPS variable vector is defined in (9.3), corresponding to the duty cycles and phase shift between bridges.

$$\mathbf{x} = [x_1 \quad x_2 \quad x_3]^T = [D_1 \quad D_2 \quad \varphi]^T \quad (9.3)$$

- Power restriction is implemented using the average transfer power expression defined above in (9.4):

$$P_{12} = \frac{4 \cdot N \cdot V_1 \cdot V_2}{\pi^3 \cdot f_{sw} \cdot L_{12}} \sum_{k=1,3,5\dots}^{2h+1} \frac{1}{k^3} \sin(k\pi D_1) \sin(k\pi D_2) \sin(k\varphi) \quad (9.4)$$

- According to the nomenclature definition, the operating limits of TPS variables are defined in (9.5) and (9.6):

$$L_b = [0 \quad 0 \quad -\pi/2] \quad (9.5)$$

$$U_b = [0.5 \quad 0.5 \quad \pi/2] \quad (9.6)$$

- Fig. 9.1 shows a graphical representation of the optimization flow: based on references for the power to be transferred and the voltages at the DC ports, the TPS values are obtained.

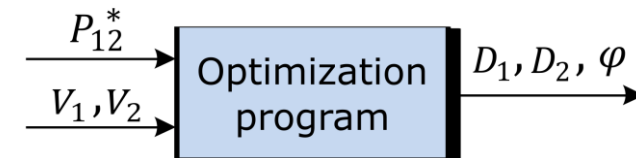


Fig. 9.1. Optimization flow representation

DAB – Optimization + Closed Loop

- Power restriction is implemented using the average transfer power expression defined above in (8.6):

$$P_{12} = \frac{4 \cdot N \cdot V_1 \cdot V_2}{\pi^3 \cdot f_{sw} \cdot L_{12}} \sum_{k=1,3,5\dots}^{2h+1} \frac{1}{k^3} \sin(k\pi D_1) \sin(k\pi D_2) \sin(k\varphi) \quad (8.6)$$

- To find a transfer function, first linearize the average power expression in (8.6) using Taylor series around an operating point:

$$\Delta P = \left. \frac{\partial P}{\partial \varphi} \right|_{\varphi_o} \cdot \Delta \varphi \quad (9.7)$$

- Then, in (9.8) the transfer function $G_{P\varphi}$ is obtained, where an operating point $\mathbf{x}_o = [D_1 \ D_2 \ \varphi_o]$ is applied to evaluate the linearization.

$$G_{P\varphi} = \frac{\Delta P}{\Delta \varphi} = \left[\frac{4 \cdot N \cdot V_1 \cdot V_2}{\pi^3 \cdot f_{sw} \cdot L_{12}} \sum_{k=1,3,5\dots}^{2h+1} \frac{1}{k^3} \sin(k\pi D_1) \sin(k\pi D_2) \cos(k\varphi_o) \cdot k \right] \Bigg|_{\mathbf{x}_o} \quad (9.8)$$

- A PI controller can then be tuned analytically or using tools such as Sisotool in MATLAB.
- In this way, a control scheme such as that shown in Fig. 9.2 can be implemented, based on the optimization control strategies discussed in [5], establishing the TPS variables with values based on an optimization module together with a closed-loop PI control that allows precise adjustment of a reference power.

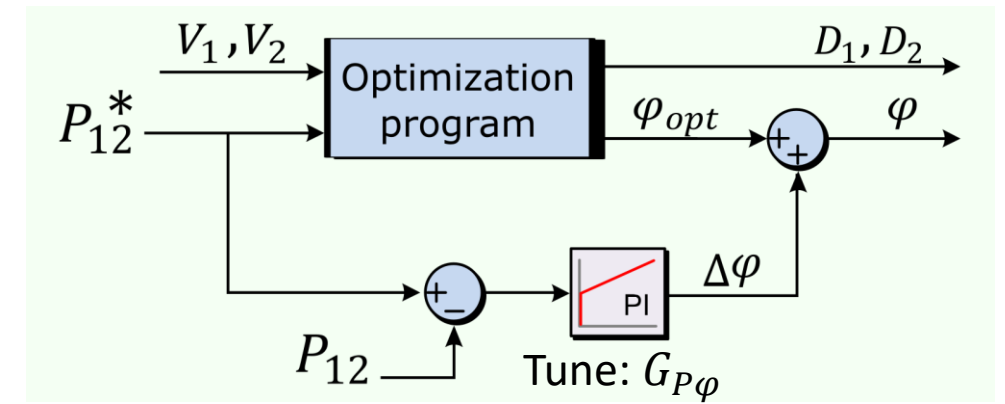


Fig. 9.2. TPS control scheme with optimization and closed loop

Section 10: Nominal and Optimal Design of the DAB

Dual Active Bridge



DAB – Design parameters

- To design a DAB, it is necessary to consider the conditions under which it will operate. It is also necessary to identify the variables or parameters to be designed.
- Known operating data:
 - ❖ Primary port voltage: V_1
 - ❖ Second port voltage range: $V_2 \in [V_{2min}, V_{2max}]$
 - ❖ Power range: $P \in [P_{min}, P_{max}]$
 - ❖ Switching frequency: f_s
- Variables to be designed
 - ❖ Transformer transformation ratio : $N = n_1/n_2$
 - ❖ Total equivalent inductance : L or L_{12}
- In accordance with the revised modulation strategies and the two studied, the DAB design will be studied in a basic form with SPS and in an optimal form with TPS.

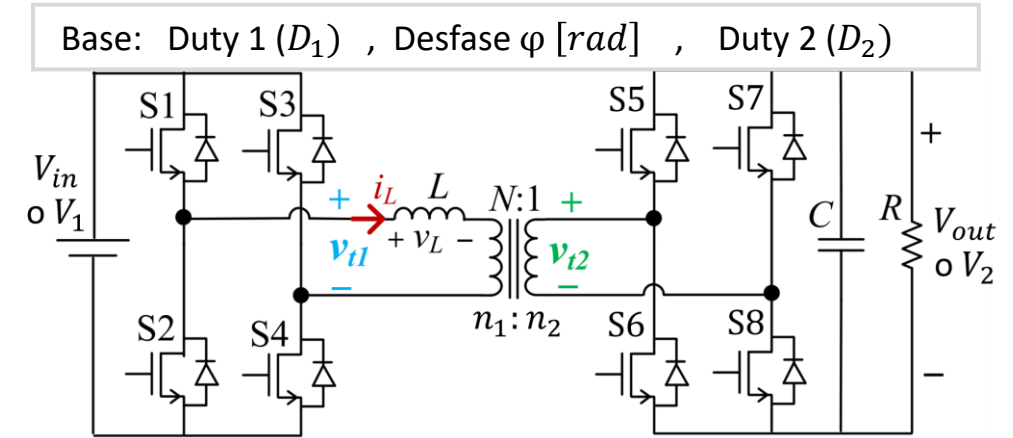


Fig. 10.1. Dual Active Non-Resonant Bridge Topology with base nomenclature [6]

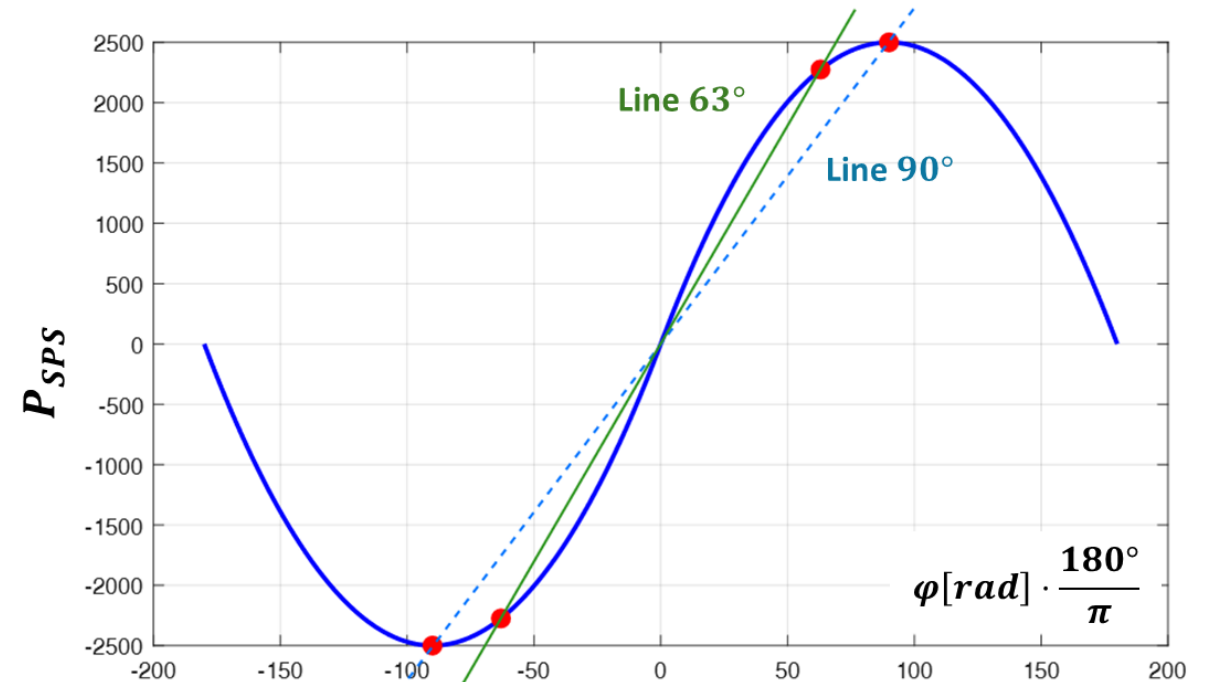


Fig. 10.2. Power transferred in the DAB

DAB – Basic design with SPS

- Operating with SPS, the power is given by (10.1) and the ratio of applied voltages M is defined in (10.2).

$$P_{SPS} = \frac{NV_1V_2}{2\pi^2 f_s L} \varphi(\pi - \varphi) \quad (10.1)$$

$$M = \frac{N \cdot V_2}{V_1} \quad (10.2)$$

- In the event of voltage variations, it is recommended to operate the DAB with values of $M > 1$ [2, 7] to achieve a greater ZVS range and lower RMS currents at nominal load. Then, the turns ratio N can be calculated in (10.3) considering the minimum secondary voltage value to ensure $M \geq 1$ throughout the range of V_2 .

$$N = \frac{V_1}{V_{2min}} \quad (10.3)$$

- Once the turns ratio has been defined, the total equivalent inductance required must be calculated. This can be determined from (10.1) and calculated in (10.4), considering the recommended angular range between 72° and 63° [2].

$$L = \left[\frac{NV_1V_2}{2\pi^2 f_s P_{max}} \varphi(\pi - \varphi) \right] \bigg|_{\varphi = \frac{63^\circ \cdot \pi}{180^\circ}} \quad (10.4)$$

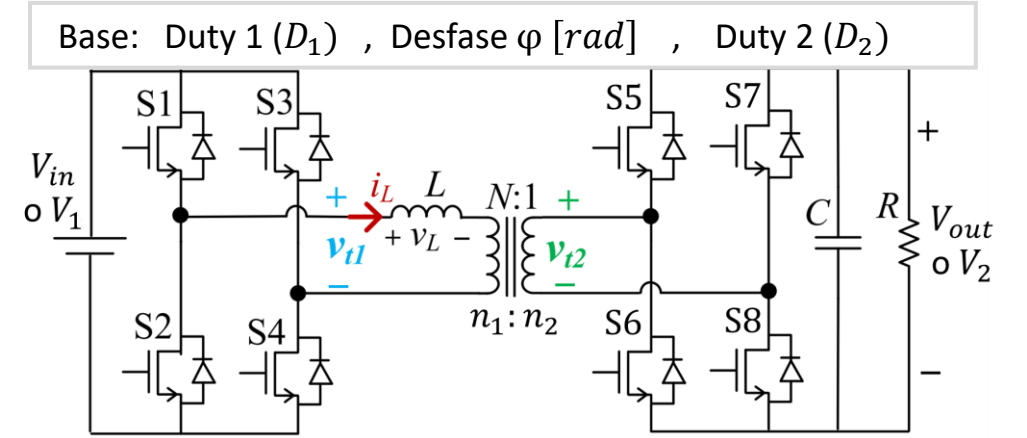


Fig. 10.1. Dual Active Non-Resonant Bridge Topology with base nomenclature [6]

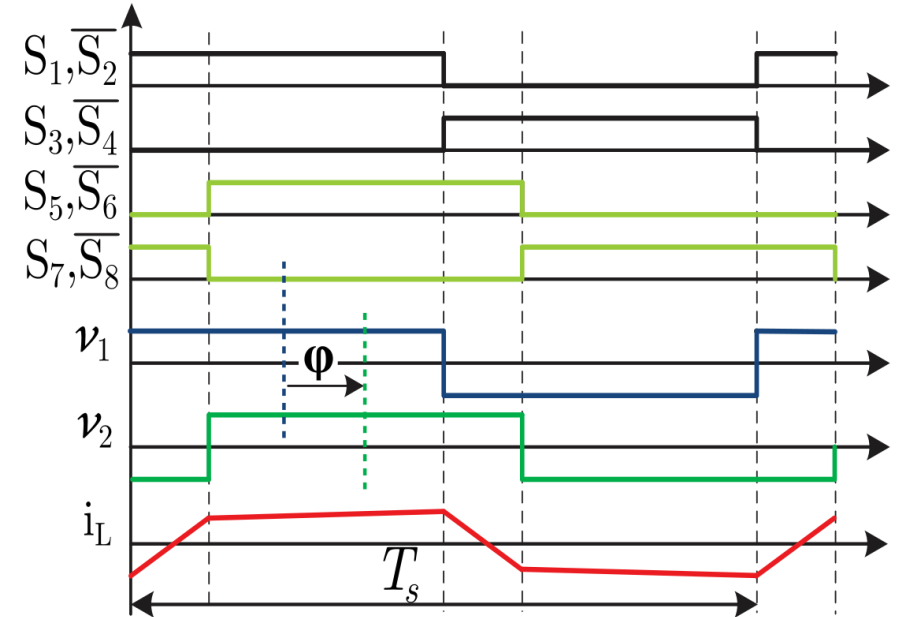
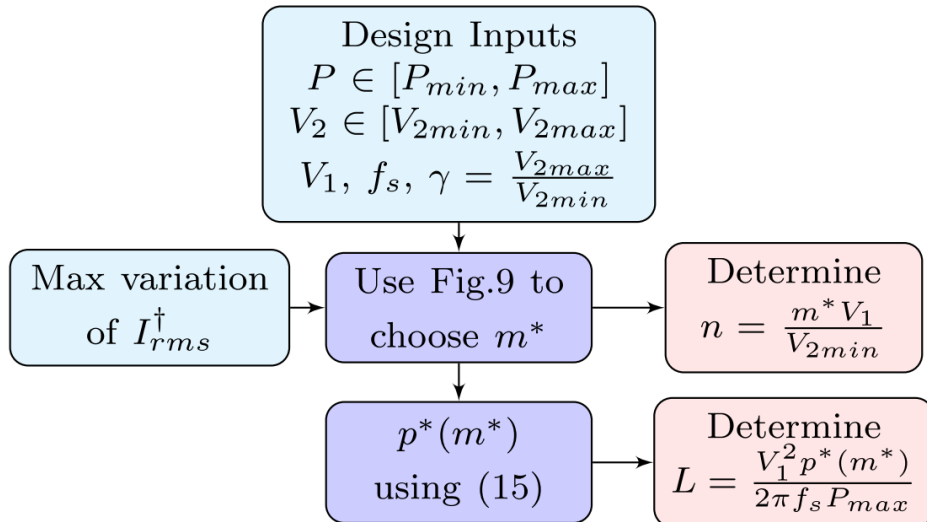


Fig. 10.3. Waveforms with SPS. Based on [5] and [6]

DAB – Optimal design with TPS

- In [7], an optimal design methodology is presented for the construction and operation of a variable-range DAB with TPS. This method considers both RMS current optimization and criteria for soft switching, adding constraints for ZVS across the entire power range operating in zones equivalent to segments operating in Case II (definition from [4]).
- The equivalence of the base nomenclature [b] with the variables of the article [4] is:
 - ❖ Turn ratio : $N = n_1/n_2$ [b] = $\mathbf{n} = \mathbf{n}_1/\mathbf{n}_2$ [7]
 - ❖ Duty cycle : D_i [b] = $\mathbf{d}_i/2$ [7]
 - ❖ Phase shift : $d_{12} = \varphi/2\pi$ [b] = $\mathbf{\delta}/4$ [7]
- The design methodology is summarized in [7] in Fig. 10.6.



[†]Percentage variation of I_{rms} due to change in V_2 at $P=P_{max}$

Fig. 10.6. Steps in the design methodology. [7]

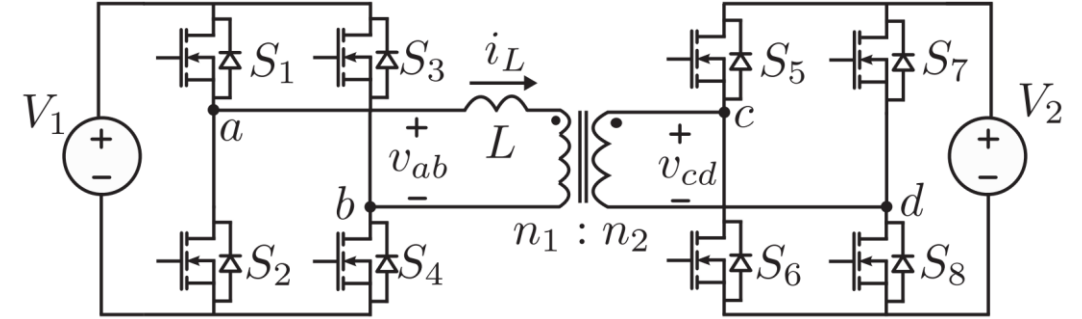


Fig. 10.4. Dual Active Bridge Topology in [7]

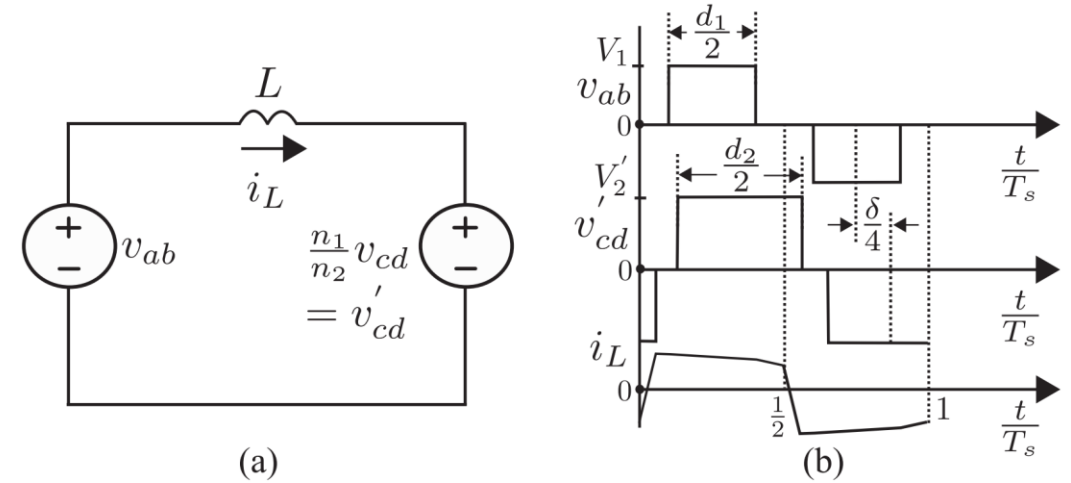


Fig. 10.5. (a) Equivalent DAB circuit (b) Waveforms with TPS. [7]

DAB – Optimal design with TPS

- The methodology [7] defines the ratio of applied voltages m in (10.5) and the ratio of secondary voltage fluctuation γ in (10.6).

$$m = \frac{n \cdot V_2}{V_1} \quad (10.5)$$

$$\gamma = \frac{V_{2max}}{V_{2min}} \quad (10.6)$$

- The principle of the analysis in [7] is to establish an initial design value for the ratio m considering the port voltages and the ratio of turns necessary to establish that m , and it is called m^* . Then, some criteria are evaluated by varying the value of m as V_2 varies, based on the parameterization of I_{rms} and P as follows:

$$I_{rms} = \frac{V_1}{2\pi f_s L} \cdot i_{rms}(m, d_1, d_2, \delta) \quad (10.7)$$

$$P = \frac{V_2}{2\pi f_s L} \cdot p(m, d_1, d_2, \delta) \quad (10.8)$$

- Fig. 10.7 shows how the RMS current of the inductor varies when the value of m changes as V_2 varies and the inductance of an initial optimal design with m^* is maintained. This leads to the **first important criterion**: since the RMS current increases rapidly for values of $m < 1$ and increases considerably more slowly for $m > 1$, it must be established that $m > 1$ is used in the variation range.

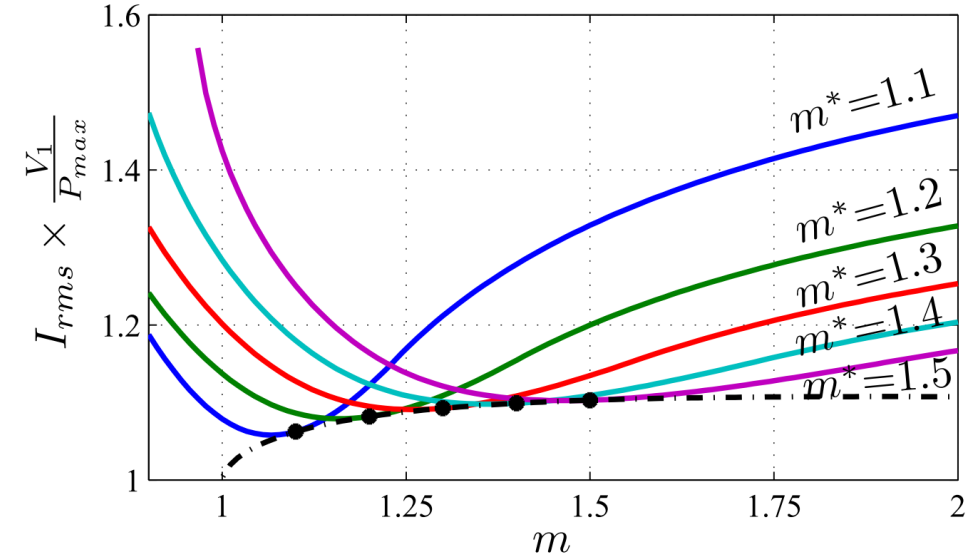


Fig. 10.7. Variation of the RMS current given the variation of V_2 for various design values of $L = L^*(m^*)$. [7]

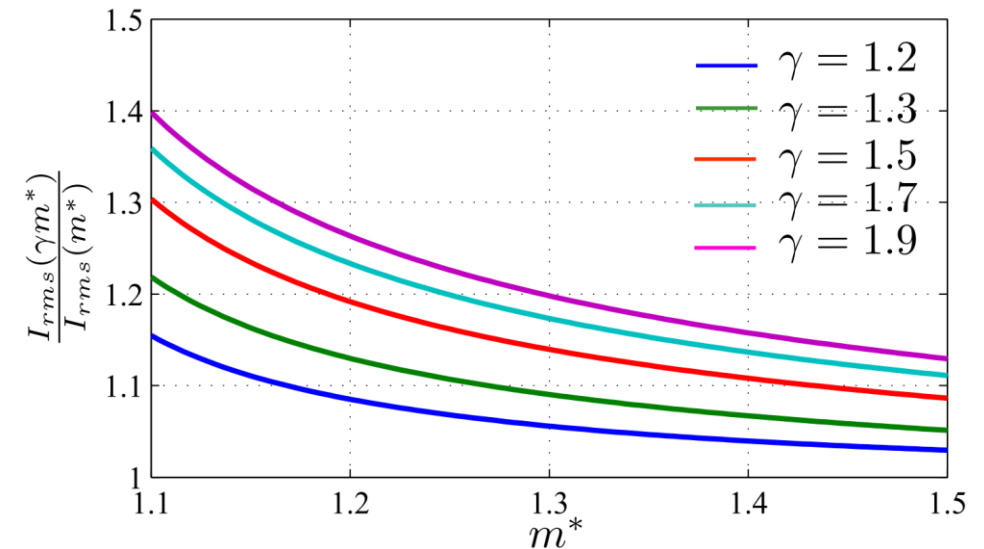


Fig. 10.8. Variation in RMS current increase given the variation in the design value m^* for various γ . [7]

DAB – Optimal design with TPS

- Fig. 10.8 shows the relation $\frac{I_{rms}(\gamma m^*)}{I_{rms}(m^*)}$, which indicates the RMS current variation depending on the design value m^* for different values of the voltage variation range of port 2 ($\gamma = \frac{V_{2max}}{V_{2min}}$). Then, as a **second criterion**, the design must set m^* considering γ and an RMS current variation tolerance.
- Interpreting Fig. 10.8, if the design m^* is small, the RMS current variation is greater at the extremes of the variation range of V_2 . Likewise, as m^* increases, this variation decreases. If, for example, with $\gamma = 1.3$, a variation of less than 10% is desired, a design of $m^* \geq 1.3$ should be chosen.
- Once the value of m^* has been determined using (10.9), the turn ratio is obtained by considering V_{2min} to achieve both criteria across the entire range:

$$n = \frac{m^* V_1}{V_{2min}} \quad (10.9)$$

- Then, using (10.10), the required inductance value is calculated. In (10.10), which has been obtained from (10.8), it is necessary to know the value of $p^*(m^*)$, defined in (10.11), which is the power per unit at the optimum design point.

$$L = \frac{p^*(m^*) \cdot V_1^2}{2\pi f_s P_{max}} \quad (10.10)$$

$$p^*(m) = -1.9m^4 + 12.6m^3 - 30.9m^2 + 34.3m - 14.07 \quad (10.11)$$

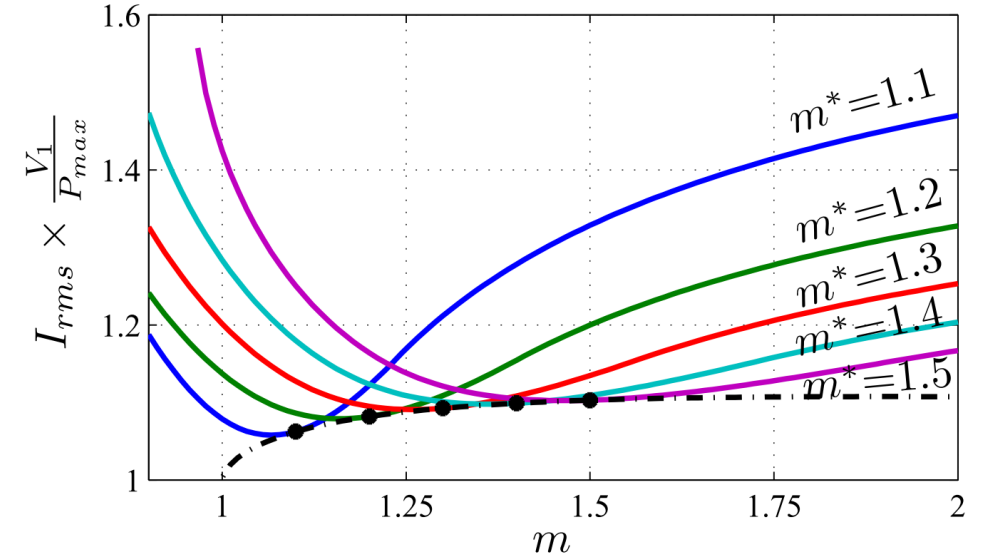


Fig. 10.7. Variation of the RMS current given the variation of V_2 for various design values of $L = L^*(m^*)$. [7]

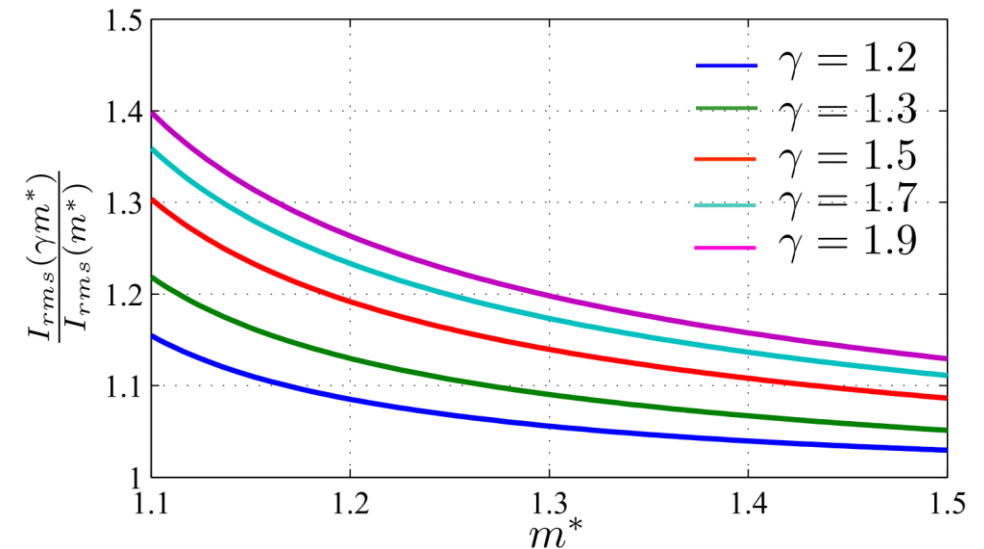


Fig. 10.8. Variation in RMS current increase given the variation in the design value m^* for various γ . [7]

Creation of LUT based on optimization with TPS

- To create an iterative program using the optimization studied, the variation of the secondary port voltage and the variation of the transferred power are considered, in order to cover a wide range of possible operating points in a DAB.
- Known data
 - ❖ Primary port voltage: V_1
 - ❖ Secondary port voltage range : $V_2 \in [V_{2min}, V_{2max}]$
 - ❖ Power range : $P \in [P_{min}, P_{max}]$
 - ❖ Switching frequency : f_s
 - ❖ Total equivalent inductance : L or L_{12}
 - ❖ Turn ratio: $N = n_1/n_2$
- Tab 10.1 shows the iterations that are formed considering the variations, generating a total number of operating points to be solved by the iteration program.

Tab 10.1. Variation Parameters for iteration

Parameter	Range	Step	Iterative variable	Iteration points	Iterative logic
Secondary port voltage	$\{V_{2min} - V_{2max}\}$ [V]	ΔV_{dc2}	i	$I = 1 + \frac{V_{2max} - V_{2min}}{\Delta V_{dc1}}$	for $i = V_{2min} : \Delta V_{dc2} : V_{2max}$ $V_{dc2} = i$; for $j = P_{min} : \Delta P : P_{max}$ $P = j$; Total puntos: total = $I \cdot J$
Power transferred from primary to secondary port	$\{P_{min} - P_{max}\}$ [W]	ΔP	j	$J = 1 + \frac{P_{max} - P_{min}}{\Delta P}$	

Creation of LUT based on optimization with TPS

- The optimization program presented above is used independently to obtain the Triple Phase Shift (TPS) modulation values for each operating point.
- When formulating the optimization problem, the restrictions for Soft Switching from [7] or [4] can be added, and even the restriction from (10.12) can be used in a simple way, which ensures Soft Switching throughout the entire operating range in case II [4] ($V_1 \geq V_2 N$ & $D_1 \leq D_2$), which is fixed with the design methodology of [7].

$$D_1 \cdot V_1 \leq D_2 \cdot V_2 \cdot N \tag{10.12}$$

- Considering Table 10.1, Figure 10.9 graphically shows how a table with optimal TPS data is generated for each operating point or combination (comb), corresponding to the IJ total combinations.

Tab 10.1. Variation Parameters for iteration

Parameter	Range	Step	Iterative variable	Iteration points	Iterative logic
Secondary port voltage	$\{V_{2min} - V_{2max}\}$ [V]	ΔV_{dc2}	i	$I = 1 + \frac{V_{2max} - V_{2min}}{\Delta V_{dc1}}$	for $i = V_{2min} : \Delta V_{dc2} : V_{2max}$ $V_{dc2} = i$; for $j = P_{min} : \Delta P : P_{max}$ $P = j$; Total puntos: total = $I \cdot J$
Power transferred from primary to secondary port	$\{P_{min} - P_{max}\}$ [W]	ΔP	j	$J = 1 + \frac{P_{max} - P_{min}}{\Delta P}$	

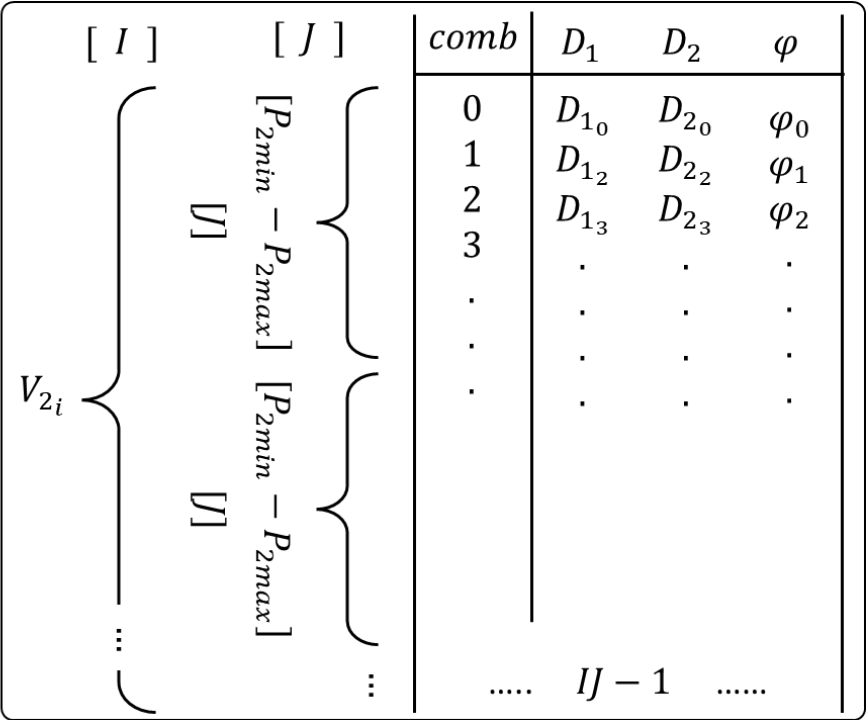


Fig. 10.9. Iteration view. There is a set of TPS values for each different operating point or combination (comb).

Creation of LUT based on optimization with TPS

		$[I]$	$[J]$	$comb$	D_1	D_2	φ
V_{2i}	$[P_{2min} - P_{2max}]$	$[U]$	$[U]$	0	D_{1_0}	D_{2_0}	φ_0
				1	D_{1_2}	D_{2_2}	φ_1
				2	D_{1_3}	D_{2_3}	φ_2
				3	.	.	.
				.	.	.	.
				.	.	.	.
				.	.	.	.
					$IJ - 1$		

- Once all the optimal TPS values for each operating point have been obtained and stored, they are stored in a single data array called DATA.

$$\Rightarrow DATA = \{ \overbrace{D_{1_0}, D_{2_0}, \varphi_0}^{comb_0}, \overbrace{D_{1_1}, D_{2_1}, \varphi_1}^{comb_1}, \dots, \overbrace{D_{1_{IJ-1}}, D_{2_{IJ-1}}, \varphi_{IJ-1}}^{comb_{IJ-1}} \}$$

- This array composed of TPS values can be viewed as a vector of $IJ \cdot 3$ positions, which can then be used to find the TPS values based on a certain secondary port voltage and required transfer power, similar to a lookup table.

$$DATA = \{ u_0, u_1, u_2, u_3, \dots, u_{IJ \cdot 3 - 1} \}$$

- The way to scan the optimal TPS data vector is based on finding the position of the vector where the TPS values for the desired combination or operating point begin. This position or Hash index is obtained using the following function. Then, as this position corresponds to the first stored value, i.e., $D_{1_{opt}}$, the next position is $D_{2_{opt}}$ and the subsequent one is φ_{opt} .

$$u_{Hash} = 3 \cdot \left[round \left(\frac{V_2^* - V_{2min}}{\Delta V_{dc2}} \right) \cdot J + round \left(\frac{P_2^* - P_{min}}{\Delta P} \right) \right] \Rightarrow \begin{aligned} D_{1_{opt}} &= DATA[u_{hash} + 0] \\ D_{2_{opt}} &= DATA[u_{hash} + 1] \\ \varphi_{opt} &= DATA[u_{hash} + 2] \end{aligned}$$

Presentation references

The 7 selected papers



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